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공학박사 학위논문

**Synthesis and Characterization of
Antibacterial Fillers and Their Application for
Polyamide Thin-film Nanocomposite Membrane and
Polypropylene Composite**

항균성 필러의 합성과 분석 그리고
폴리아마이드 박막 나노복합막과
폴리프로필렌 복합재료에의 응용

2024년 8월

서울대학교 대학원

화학생물공학부

서 희 란

**Synthesis and Characterization of
Antibacterial Fillers and Their Application for
Polyamide Thin-film Nanocomposite Membrane and
Polypropylene Composite**

by

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**School of Chemical and Biological Engineering
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항균성 필러의 합성과 분석 그리고
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지도 교수 이 중 찬 박사

이 논문을 공학박사 학위논문으로 제출함
2024년 8월

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Abstract

Synthesis and Characterization of Antibacterial Fillers and Their Application for Polyamide Thin-film Nanocomposite Membrane and Polypropylene Composite

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This study presents the synthesis and characterization of antibacterial fillers that exhibit high compatibility with the polymer matrix and strong antibacterial activities. It investigated enhanced antibacterial properties and various changes in the properties of each polymer composite resulting from the incorporation of antibacterial fillers into polyamide (PA) thin-film composite membranes and polypropylene (PP) composites. Firstly, cellulose nanocrystals (CNC) were modified using 3-pentadecylphenol (PDP), a cardanol-derived material, to prepare amphiphilic functionalized CNC (f-CNC). These f-CNCs were then dispersed as nanofillers in an organic solvent during the interfacial polymerization process to fabricate PA-f-CNC

composite membranes. The introduced f-CNCs manipulated the monomer diffusion process during interfacial polymerization and physically inhibited excessive growth of the PA layer, significantly reducing the thickness and permeation resistance of the PA and thereby enhancing the water permeability of the membrane. Despite the incorporation of f-CNCs, the PA membrane maintained its selectivity and showed stable water permeability and salt rejection over long-term operation. This demonstrates that f-CNCs were uniformly distributed within the PA matrix without causing any deterioration in separation performance due to agglomeration of fillers. Additionally, the simultaneous introduction of CNC, which can puncture bacterial cell walls with its rigid and sharp structure, and PDP, characterized by long alkyl chains that provide contact-killing effects, endowed the PA-f-CNC composite membrane surface with strong antibacterial properties. When bacteria or organic foulants were added into feed solution, the PA-f-CNC membrane exhibited less decline in water permeability during fouling cycle and higher recovery after washing cycle compared to the neat PA membrane. In summary, the introduction of f-CNC significantly enhanced the permeability, antibacterial properties, and fouling resistance of the PA membrane simultaneously.

Secondly, zinc silicate glass powder (ZS), which exhibits antibacterial activity through the zinc ion release, was modified using a silane coupling

agent featuring an alkyl chain of # carbons and quaternary ammonium groups, resulting in the synthesis of ZS#AC. As-prepared ZS#AC was then incorporated into PP to fabricate the PZ#A_x composites (where *x* represents the mass ratio of the filler to PP). The alkyl chains in ZS#AC improved the dispersion stability of the fillers within the PP matrix, facilitating the release of zinc ions from the PP composite film and thereby enhancing its antibacterial properties. The quaternary ammonium groups strengthened and sustained the antibacterial activity of PP composites through the contact-killing effect even after the consumption of zinc ions. In addition, the introduction of ZS#AC, which has rigid property and high compatibility with the PP matrix, contributed to the improvement of the mechanical and thermal properties of the PP. In summary, the incorporation of ZS#AC simultaneously enhanced the antibacterial properties, mechanical properties, and thermal stability of the PP composites.

Keywords: Polymer composite, Antibacterial filler, Polyamide thin-film nanocomposite membrane, Polypropylene composite, Cellulose nanocrystal, Zinc silicate glass powder, Filler modification, Filler dispersion stability

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Chapter 1

Introduction

1.1. Antibacterial polymer composites

Polymer composites containing nanofillers have been extensively utilized across various fields due to their enhanced properties such as mechanical, thermal, chemical, and electrical properties [1-3]. Recently, with growing concerns regarding bacterial infections, numerous studies have focused on polymer composite materials with antibacterial properties [4]. Antibacterial polymer composites can be prepared by dispersing antibacterial agents within a polymer matrix and have been applied in fields such as medical materials, packaging, and separation membranes [5-7]. These antibacterial nanofillers can prevent bacterial proliferation on polymer surfaces either by killing the bacteria or inhibiting their attachment.

Antibacterial nanofillers can be classified into two types: inorganic and organic. Inorganic antibacterial nanofillers, specifically metal oxide such as silver nanoparticles, copper nanoparticles, zinc oxides, and titanium dioxides, exhibit strong bactericidal effects [8,9]. Organic antibacterial nanofillers include antimicrobial agents such as quaternary ammonium compounds and derivatives of nature-derived substances known for their bactericidal properties, such as cardanol and citronellol [10,11]. These nanofillers can impart antibacterial activities to polymer composites through various mechanisms. They may disrupt cellular metabolism via metal ions,

induce DNA damages by releasing reactive oxygen species, or destroy bacterial cell walls through direct contact between the antimicrobial agents and the bacteria [12].

To enhance the antibacterial activity of polymer composites without compromising their mechanical stability, the antibacterial fillers must exhibit high compatibility with the polymer matrix [13]. Furthermore, it is crucial that these antibacterial polymer composites are biocompatible, non-cytotoxic, and environmentally safe [14]. The development of such polymer composites will facilitate the safe application of polymer materials across various fields, alleviating concerns about bacterial infections.

1.2. Antibacterial polymer membranes for water treatment

Polymer membranes are widely used in various fields such as microfiltration (MF), ultrafiltration(UF), nanofiltration (NF), and reverse osmosis (RO) because their pore size and surface properties can be easily controlled [15,16]. This allows for the selective separation of various particles, including suspended solids, viruses, bacteria, and ions, based on sieving, Donnan-exclusion, or solution-diffusion mechanisms [17,18]. Polymer membranes provide a cost-effective and easily manufacturable alternative to inorganic membranes. However, polymer membranes have the disadvantage of being prone to surface fouling due to their inherent hydrophobicity [19]. Specifically, biofouling, which results from the bacterial growth and biofilm formation on the membrane surface or inside the pores, significantly reduces the membrane separation performance and shortens the life of membrane [20].

Therefore, numerous studies have focused on polymer nanocomposite membranes containing antibacterial nanofillers to mitigate the attachment of growth of bacteria [21,22]. Silver nanoparticles (AgNP), having strong bactericidal activities, are commonly used as nanofillers for the fabrication of polymeric membranes such as cellulose acetate [23], polysulfone [24], polyethersulfone [25], and polyamide [26]. They demonstrated anti-

biofouling properties against bacteria such as *E. coli*, *S. aureus*, and *P. aeruginosa*. Graphene oxide (GO) [27] and carbon nanotubes (CNT) [28], which exhibit antibacterial activity through puncturing or contact-killing, are employed in the preparation of antibacterial nanocomposite membranes. Metal-organic frameworks (MOF) [29], which release antibacterial metal ions, and quaternary ammonium compounds [30], which exert bactericidal activities through interactions with bacterial cell walls, are also commonly used as antibacterial fillers for polymer water treatment membranes. These diverse types of antibacterial nanofillers have broadened the application of polymer membranes by preventing membrane performance degradation caused by bacterial fouling.

1.3. Antibacterial polypropylene composites

Polypropylene (PP) is one of the most widely used thermoplastics due to its excellent mechanical properties, optical transparency, thermal and chemical stability, and cost efficiency [31]. PP can be easily molded into desired shapes, making it versatile for various household items and industrial parts. However, the surface of PP is prone to bacterial growth because it has inherent hydrophobic properties and lacks any antibacterial moieties [32]. Therefore, antibacterial PP composites have been developed by incorporating the fillers containing metals such as silver, copper, and zinc. Jo et al. [33] developed a PP nanocomposite film suitable for food packaging by incorporating silver nanoparticles, demonstrating over 99.9% antibacterial activity against *E. coli* and *S. aureus*. Silvestre et al. [34] presented a photodegradable PP composite with zinc oxide, showing more than 99.9% antibacterial property against *E. coli*. Jiang et al. [35] prepared a PP composite that exhibits antibacterial properties against four types of bacteria by introducing nanocopper. They reported that the controlled release of copper ions does not adversely affect human health, allowing copper ions to be utilized safely as antibacterial fillers.

However, these nanofillers tend to agglomerate within hydrophobic PP, decreasing the mechanical properties of PP and diminishing the reinforcing

effect of the fillers [36]. Therefore, to effectively impart antibacterial properties without compromising the inherent characteristics of PP, it is essential to introduce functional groups that can enhance compatibility between the fillers and the PP matrix.

1.4. Motivation

Based on the need for antibacterial polymer composites and a comprehensive understanding of various fillers, I aimed to incorporate these fillers into polyamide thin-film composite membrane and polypropylene. Specifically, to maximize the effectiveness of the antimicrobial fillers and to prevent performance degradation due to aggregated filler, these fillers were modified using moieties that exhibit high compatibility with the polymer matrix. Since the introduced filler affects not only the antibacterial properties but also other properties of the polymer composite, these effects were also thoroughly investigated.

To fabricate the antibacterial polyamide membrane, rod-like cellulose nanocrystal (CNC), which can puncture bacterial cell walls through its sharp and stiff properties, was utilized as a filler. To enhance the dispersion stability of CNC in the organic solvent used for polyamide synthesis process and to introduce additional antibacterial moieties, CNC was modified with 3-pentadecylphenol (PDP), a bactericidal material derived from cardanol. Subsequently, functionalized CNC (f-CNC) was introduced as an antibacterial filler in the polyamide membrane.

For the fabrication of antibacterial polypropylene composites, zinc silicate glass powder (ZS) was utilized, which incorporated zinc oxide into

the silica network to reduce cytotoxicity and minimize photocatalytic effect. The ZS was modified using a silane coupling agent to simultaneously introduce alkyl chain groups, which enable the hydrophilic ZS to be dispersed evenly within the polypropylene matrix, and a quaternary ammonium groups that imparts strong and sustainable antibacterial properties.

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Chapter 2

**Interfacial diffusion manipulation by amphiphilic
cellulose nanocrystal having antibacterial property
for asymmetric polyamide layer**

2.1. Introduction

Thin-film composite (TFC) reverse osmosis (RO) membranes, fabricated by interfacial polymerization (IP) process, have been widely used in the desalination processes to address the global water crisis [1-4]. Typically, TFC membranes comprise two distinct configurations: a thick and loose porous substrate providing structural stability under high pressure, and a thin and dense polyamide (PA) selective layer [5]. The separation performance of a TFC membrane is significantly influenced by the structure of the PA layer, which serves to separate water and solutes [6,7]. Although current TFC membranes exhibit acceptable performance, they have reached the permeation-selectivity upper-bound and still suffer from membrane performance degradation upon filtration [8-10]. The fouling occurs when organic matters and/or microorganisms attach and accumulate on the membrane surface upon filtration, significantly lowering membrane lifetime [11]. With these issues, fundamental investigations and practical approaches to improve water permeation, salt rejection, and antifouling/antimicrobial properties altogether remain major challenges.

Thin-film nanocomposite (TFN) membranes, where nanofillers are typically integrated into the PA selective layer, stand out as a breakthrough solution for the challenges, offering improved separation performance and

antifouling properties simultaneously [12-14]. Various nanofillers, including carbon nanotube (CNT) [15,16], graphene oxide (GO) [17], metal organic framework (MOF) [18,19], and titanium dioxide [20,21], have been employed to facilitate the formation of advanced PA layers. Notably, cellulose nanocrystal (CNC) is gaining significant attention as a sustainable nanofiller material, easily obtained by acid hydrolysis of natural resources (e.g., wood and cotton) [22]. CNC possesses high hydrophilicity, non-toxicity, excellent mechanical stability, and a large surface area, making it attractive for improving water treatment membranes [23-26]. Furthermore, a number of hydroxy groups on the surface facilitate easy functionalization through simple chemical reactions [27,28]. However, in most cases, hydrophilic nanofillers including CNC are dispersed in an aqueous solution rather than an organic solution prior to IP, resulting in a significant portion of the nanofillers being washed away during the IP process [29,30]. Although, F. Asempour et al. reported the TFN membranes with improved water permeability by introducing unmodified CNC into the organic solvent layer, there was a decrease in salt rejection, a typical trade-off phenomenon due to incomplete dispersion [31]. Therefore, we believe that hydrophobic surface modification of CNC would provide additional advantages when incorporating CNC into the organic phases. Some of hydrophilic nanofillers, such as GO [32] and MOF [33], have been functionalized with long alkyl

chains to facilitate their dispersion in an organic phase during the IP process. However, while acetylation has been conducted on CNC [34], there has been no study on utilizing CNC modified with long alkyl chains as a nanofiller in the organic solvent layer for the advanced TFN membranes. Furthermore, there is still much to learn about how amphiphilic modified nanofillers affect the dynamics of diffusion-based IP process.

Therefore, in this study, we attempted to improve the dispersion stability of CNC in organic solvents through the modification with long alkyl chain groups. Simultaneously, we closely examine how amphiphilic functionalized cellulose nanocrystal (f-CNC), synthesized by introducing oleophilic long alkyl chain groups into hydrophilic CNC, manipulates interfacial diffusion in the IP process. We comprehensively investigated the influence of introducing f-CNC on interfacial tension and monomer diffusion during the IP process, as well as the subsequent changes in morphology and separation performance of the membranes. To the best of our knowledge, this is the first work that comprehensively investigate the effects of amphiphilic nanofiller dispersed in an organic phase on IP dynamics. For the modification of CNC, 3-pentadecylphenol (PDP), a cardanol-derived material featuring a long alkyl chain group and exhibiting antibacterial properties, was employed. Notably, this marks the first attempt to modify CNC using natural materials for application in a TFN membrane.

An additional advantage of employing CNC and PDP to prepare f-CNC is the utilization of abundant bio-derived materials, contributing to the sustainability of this approach [24,35,36].

Here, we show that by incorporating amphiphilic f-CNCs into an organic phase during the IP, the PA layer with thin asymmetric morphology containing many voids can be produced, enhancing water permeability while retaining salt selectivity [7,37,38]. This distinct architecture was achieved by the influence of the amphiphilic f-CNC located near the water/heptane interface: (i) the f-CNC lowers the interfacial tension, inducing the formation of a thin and dense basal PA layer and (ii) it acts as a physical barrier, confining the PA growth in the upper part (Figure 2.1). Building on our recent demonstrations that enhanced water permeability without trade-off by incorporating polymer surfactant or nanofillers in the aqueous phase [15,17,39,40], we further revealed the impact of amphiphilic nanofillers in the organic phase on IP dynamics. In addition, we demonstrate that this nanocomposite membrane with f-CNC can exhibit improved antifouling and antibacterial properties attributed to the intrinsic properties of CNC and the bactericidal activity of the PDP [28]. The effect of hydrophilic CNC on the fouling resistance and antibacterial properties of TFN membranes has been investigated, but the effect of amphiphilic modified CNC on the fouling resistance and antibacterial properties of the

membrane was investigated for the first time in this study.

2.2. Experimental

2.2.1. Materials

Freeze-dried CNCs (length: 50–400 nm, diameter: 10–20 nm, zeta-potential: -36.5 ± 0.35 mV, and counter ion: Na^+) were obtained from Cellulose Lab Co., Ltd. (USA). 4-Oxo-4-(3-pentadecylphenoxy)butanoic acid (PDPBA) was synthesized by the reaction between 3-pentadecylphenol (PDP) and succinic acid (Figure 2.2) [28]. Polysulfone (PSf) membrane (US100P, Microdyn-Nadir, Germany) with molecular weight cut-off (MWCO) of 100 kDa was used as a support membrane. m-Phenylenediamine (MPD, 99%, Sigma-Aldrich), trimesoyl chloride (TMC, 98%, Sigma-Aldrich), p-toluenesulfonic acid (PTSA, 98.5%, TCI), sodium dodecyl sulfate (SDS, 85%, TCI), triethylamine (TEA, 99%, TCI), SytoTM 9 (Life Technologies Corporation), and propidium iodide (PI, Sigma-Aldrich) were used as received. *Escherichia coli* (*E. coli*, ATCC 8739) was received from American Type Culture Collection (ATCC). DifcoTM Luria-Bertani (LB) Broth and Agar were supplied from Becton, Dickinson and Company (USA). Deionized (DI) water (resistivity, 18.3 m Ω ·cm) was obtained through a water purification system (Human Power I+, Human Corporation, Republic of Korea). All other chemicals were used as received from general chemical vendors unless otherwise specified.

2.2.2. Functionalization of CNC

CNC was functionalized by the esterification between hydroxy groups of CNC and carboxylic acid groups of PDPBA. General procedure was followed by previously reported method [28]. 5.0 g of CNC, 19.2 g of PDPBA, 0.08 g of PTSA, and a magnetic stir bar were placed in a 250 mL round-bottom flask, followed by adding 150.0 mL of anhydrous toluene. The mixture was sonicated for 30 min and heated by immersing in an oil bath at 120 °C with stirring. After 24 h, the crude solution was diluted with excess amount of hexane, and the solid was obtained by vacuum-assisted filtration using polytetrafluoroethylene (PTFE) filter (pore size, 0.2 μm). The remained unreacted chemicals were removed by continuously washing the filtered solid with chloroform and hexane. The final solid, f-CNC, was dried in the 30 °C vacuum oven overnight and stored in a desiccator prior to use.

2.2.3. Fabrication of thin-film composite (TFC) and thin-film nanocomposite (TFN) with f-CNC membranes

In this study, the membrane without filler was defined as thin-film composite (TFC) membrane, and the membrane with nanofillers was defined as thin-film nanocomposite (TFN) membrane. Both TFC and TFN

membranes were prepared through interfacial polymerization (IP) on the polysulfone (PSf) non-woven support membrane. IP was performed using previously reported method, except for the addition of nanofillers. The support membrane was placed in an aqueous solution containing MPD (2.0 w/v%), SDS (0.1 w/v%), and TEA (0.1 w/v%), and excess aqueous solution on the membrane surface was removed after 5 min. An organic solution (TMC, 0.1 w/v% in heptane) was poured on the membrane inducing IP. In case of TFN membrane, f-CNC was dispersed in TMC/heptane solution with pre-determined concentrations (0.01–0.05 w/v%). The interfacial reaction was conducted for 1 min, excess heptane solution was discarded, and the membrane was heated at 85 °C in an oven for 5 min. Finally, the as-prepared membrane was washed with DI water several times and stored in DI water prior to analyze or test. In addition, other membranes were prepared for comparison using the same procedure except for following modification: CNC (not functionalized, 0.02 w/v% in water) or PDPBA (0.02 w/v% in heptane).

2.2.4. Membrane filtration test

Separation performance of membranes was evaluated by the lab-scale cross-flow RO membrane filtration test unit. The effective membrane area was $4.1 \times 6.1 \text{ cm}^2$ with the 0.3 cm of channel height. The operating pressure

was 15.5 bar and cross-flow velocity was about 0.8 L/min at 25 °C. The membranes were pre-stabilized by the filtration of DI water for 1.5 h at 15.5 bar. The water flux was evaluated by weighing the permeate for 10 min using DI water and calculated by Eq. (1):

$$J_w (\text{L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}) = \Delta V / (A \times \Delta t \times P) \quad (1)$$

where ΔV is the volume of permeate (L), A is the effective area of membrane (m^2), Δt is the time taken to collect the permeate (h), and P is the operating pressure (bar).

The salt rejection (R) was estimated using 2000 ppm NaCl feed solution and calculated by Eq. (2):

$$R (\%) = (1 - C_p / C_f) \times 100 \quad (2)$$

where C_p is the salt concentration of the permeate, and C_f is the salt concentration of the feed. The salt concentration was evaluated by measuring conductivity of the permeate and feed solution using a conductivity meter (HI 5321, Hanna instruments, USA). All water flux and salt rejection values displayed in this study are the average values that calculated from at least three measurements on different membranes prepared under the same condition.

Antifouling properties of the membranes were also evaluated by the lab-scale cross-flow RO membrane filtration test unit under the same conditions as the separation performance evaluation except for the feed solution: the

feed solution contains BSA (1000 ppm, as model foulant) and NaCl (2000 ppm). Membranes were compacted with DI water for 1.5 h and 2000 ppm NaCl solution for 0.5 h to stabilize the water flux (15.5 bar and 25 °C). Then, the water permeation values were evaluated in every 10 min, and the following fouling/releasing cycles were repeated for 3 times: (i) the feed solution was replaced with BSA/NaCl solution, and run for 6 h (i.e., fouling cycle), (ii) the feed solution was replaced with DI water, and run for 2 h (i.e., washing cycle), and (iii) the feed solution was replaced with NaCl solution, and run for 2 h (i.e., recovery cycle).

To estimate the antifouling properties of the membranes, the flux decline ratio (*FDR*) and flux recovery ratio (*FRR*) were calculated by Eq. (3) and Eq. (4):

$$FDR (\%) = (1 - J_p / J_i) \times 100 \quad (3)$$

$$FRR (\%) = (J_r / J_i) \times 100 \quad (4)$$

where J_i was the initial water flux, J_p was the water flux measured after each BSA fouling process, and J_r was the water flux after DI water washing process.

The antibiofouling properties of the membranes were also evaluated by filtration of the solution containing microorganism (*E. coli*) (10^6 CFU/ml in a PBS solution, pH 7.4). The water flux was measured in every 1 h, and the biofouling/releasing cycles were repeated for 2 times. In each cycle,

biofouling was induced for 20 h using the bacterial solution and releasing was carried out for 2 h using DI water. During the biofouling experiment, the temperature of feed solution was kept constant at 30 °C and the cross-flow velocity was maintained at about 0.4 L/min. Note that relatively high feed temperature and low cross-flow velocity were employed to facilitate microorganism attachment.

2.2.5. Antibacterial properties evaluation

The antibacterial properties of membranes were evaluated by a plate counting method. The membranes (1.0 cm × 1.0 cm) were kept in 1 mL of suspension of *E. coli* (10^5 CFU/mL) for 24 h at 25 °C. After discarding the bacteria suspension, the membranes were immersed in 1.0 mL of 0.9 wt% NaCl solution. Then, bath sonication was conducted for 10 min to detach the bacteria from surface of the membranes. The *E. coli* suspension was diluted, and 0.1 mL of diluent was spread on the agar plates. Subsequently, the agar plates were incubated at 37 °C for 18 h and bacterial colonies were counted. Antibacterial properties of each membrane were measured at least three times. Antibacterial properties were calculated by Eq. (5):

$$\text{Antibacterial property (\%)} = (N_B - N_S) / N_B \times 100 \quad (5)$$

where N_B is the number of bacterial colonies of blank and N_S is the number of bacterial colonies of the tested membrane samples.

Additionally, live and dead bacteria cell were observed by CLSM. The membranes (2.0 cm × 2.0 cm) were kept in 5.0 mL of suspension of *E. coli* (10^8 CFU/mL) for 24 h at 25 °C. Then, the membranes were washed with 0.9 wt% NaCl solution and stained by fluorescent dye mixture of 0.02 mM of SytoTM 9 and 0.001 mM of PI. SytoTM 9 is the green nucleic acid dye for all cells, and PI is the red nucleic dye for dead cells which have the destroyed cell walls. In CLSM images, the live cells were showed green color and the dead cells showed red color.

2.2.6. Characterizations

The synthesis of PDPBA was confirmed by ¹H-NMR spectroscopy (Advance III 400 FT-NMR, Bruker, USA). The chemical structures of CNC, PDPBA, and f-CNC were characterized by Fourier-transform infrared spectroscopy (FT-IR, Nicolet 6700, Thermo Scientific, USA) in transmission mode with a 4 cm⁻¹ resolution. Thermogravimetric analysis (TGA, Q50, TA Instrument, USA) was conducted on CNC, PDPBA, and f-CNC in the temperature range of 100–700 °C with a heating rate of 5 °C min⁻¹ under N₂ atmosphere. The surface compositions of nanofillers and membranes were characterized by X-ray photoelectron spectroscopy (XPS, AXIS-His, KRATOS, UK) with Al K α (1486.6 eV) radiation source. The morphologies of CNC and f-CNC were observed by transmission electron

microscopy (TEM, JEM1010, JEOL, Japan) with an accelerating voltage of 80 kV. 10.0 mg of CNC or f-CNC was dispersed in 10 mL of water and heptane, respectively, and the suspension was sonicated at least 2 h. Then, a drop of the suspension was deposited on a copper TEM grid. As-prepared grid was dried and stored in 30 °C vacuum oven. The apparent particle sizes of CNC in water and f-CNC in heptane were investigated using dynamic light scattering spectrophotometer (DLS, ELS Z-neo, Otsuka Electronics, Japan). The surface and cross-sectional morphologies of TFC and TFN membranes were examined by field-emission scanning electron microscopy (FE-SEM, SUPRA 55VP, Carl Zeiss, Germany). The cross-sectional morphologies and thickness of TFC and TFN membranes were observed by energy-filtering transmission electron microscope (EF-TEM, LIBRA 120, Carl Zeiss, Germany) with an accelerating voltage of 120 kV. The membrane samples were embedded in Spurr's resin and cured for 48 h. The ultrathin sections (70 nm) were cut by a ultramicrotome (EM UC7, Leica, Germany) and collected on copper grids. To examine the effect of f-CNC on the MPD diffusion rate, the concentration of MPD in the organic solution over time was measured by UV-Vis spectrophotometer (U-5100, Hitachi, Japan) with a 1.0 cm quartz cuvette at room temperature [41]. First, 0.5 mL of 2.0 w/v% MPD aqueous solution was placed in a cuvette, then 2 mL of heptane solution containing f-CNC was added, and UV absorbance was

recorded at 295 nm. Interfacial tension of water/heptane interface was computed by pendant drop mode of drop-shape analyzer (DSA 25, KRUSS, Germany) at room temperature. The drop phase was the DI water and the surrounding phase was the heptane, and the CNC, f-CNC, and surfactant (SDS) was dispersed at a specific concentration. The Young-Laplace equation was used as the fitting method. Also, the surface hydrophilicity and surface energy of the membranes were investigated by contact angle measurement using sessile drop mode of drop-shape analyzer (DSA 25, KRUSS, Germany). To determine the surface energies, DI water was used as polar solvent and di-iodomethane was used as non-polar solvent. The images of heptane droplets in water were observed by inverted light microscope (Axio Observer Z1, Carl Zeiss, Germany). The surface roughness of as-prepared membranes was analyzed by atomic force microscopy (AFM, NX-10, Park systems, Republic of Korea). The AFM images were observed in $5.0\ \mu\text{m} \times 5.0\ \mu\text{m}$ using the non-contact mode in atmosphere at a scan rate of 0.3 Hz.

2.3. Results and Discussion

2.3.1. Dispersion of f-CNCs in an organic solution

The CNC surface was modified by the esterification with PDPBA (Figure 2.3a); note that PDPBA is derived from renewable cashew nutshell oil, enabling sustainable development, and the preparation of f-CNC is previously reported [28,36]. We expected this modification amends the surface nature of CNC from hydrophilic to oleophilic due to the incorporation of long alkyl chains in PDPBA moiety, thereby improving CNC dispersion in organic solution (i.e., heptane) that used in IP process. The dispersion of the nanofiller in liquid media during IP process is one of the most important factors for the successful preparation of nanocomposite membranes [13,33,40,42]. The functionalization of CNC was confirmed by Fourier-transform infrared spectroscopy (FT-IR), thermal gravimetric analysis (TGA), and X-ray photoelectron microscopy (XPS) analyses (Figures 2.4-2.6 and Table 2.1). Briefly, the decrease in O–H and increase in C=O stretching peaks after the functionalization in FT-IR spectra (Figure 2.4) confirmed the successful chemical reaction (i.e., esterification) between CNC and PDPBA, and the TGA data suggested the weight fraction of PDPBA in f-CNC is ~ 50 % (Figure 2.5).

Figure 2.3b shows the TEM images of CNC (pre-dispersed in water) and f-CNC (pre-dispersed in heptane). Both were well dispersed in the respective solvents. A little aggregation in f-CNC is probably due to its amphiphilic nature composed of hydrophobic pentadecyl moieties on the surface and hydrophilic cellulose bodies. Although the width of the CNCs slightly increased and more agglomerated regions were observed after modification, both CNC materials showed similar appearances, with rod-like structures and interconnected network formations [28,43-45]. Laser irradiation to the f-CNC/heptane solution was carried out: the observation of the Tyndall effects is one of the most intuitive methods for visualization of dispersion stability (Figure 2.3c). All f-CNC suspensions showed distinct Tyndall effects, i.e., very fine light scattering in a suspension, indicating that the f-CNC has high dispersion stability in heptane [46,47]. The thicker and more scattered laser at high f-CNC concentration is due to some aggregated particles. On the contrary, non-modified CNC solution showed no laser scattering, suggesting the poor dispersion stability of CNC in heptane. In addition, the apparent particle size of f-CNC dispersed in heptane was similar to that of CNC dispersed in water (Table 2.2), suggesting that f-CNC is well-dispersed in heptane given that CNC is known to be well-dispersed in water [34]. This could be supported by the gradual increase in UV-Vis absorbance with increasing f-CNC concentration in heptane (Figure 2.7).

2.3.2. Separation performances of TFN membranes with f-CNCs

PA thin film nano-composite (TFN) membranes were prepared by the typical IP process [2,5,12] except for using f-CNC/TMC/heptane solution as an organic phase: PA-f-CNC_# indicates the TFN membrane prepared with # w/v% f-CNC/heptane solution. Figure 2.8 shows water permeability and salt rejection values of the TFN membranes. In contrast to our expectation that water permeability should be improved over a certain amount of f-CNC loadings (*e.g.*, > 0.1 w/v%) given that previously reported threshold of CNC loading is ~ 0.1 w/v% in solution or larger for achieving high water permeability and salt selectivity (Table 2.3) [25,31,34,48,49], the TFN membranes in this study exhibited a dramatic increase in water permeability at very low f-CNC loadings (~ 0.02 w/v%) while retaining high salt rejection values. This indicates that employing f-CNC offers an economical and practical advantages because the separation performance of the membrane can be enhanced with very low nanofiller loadings. At higher f-CNC loadings over 0.03 w/v% in heptane, the salt rejection values decreased without dramatic increase in water permeability, presumably due to the generation of defects in a PA layer, originating from the aggregated f-CNC microparticles (Figure 2.9) [50,51]. In this regard, we concluded that the optimal f-CNC loading is ~ 0.02 w/v% in heptane, leading to $\sim 44\%$

water permeability improvement with a minimal salt rejection sacrifice. The ratio of water (A) and salt (B) permeability coefficients (A/B) represents the water-salt selectivity of the membrane [52]. In comparison to the neat PA, PA-f-CNC_0.02 exhibited increased values for both water permeability coefficient (A) and water-salt selectivity (A/B), overcoming the trade-off relationship (Table 2.4 and Figure 2.10). In addition, f-CNC and PA could have good compatibility given the stable membrane performance of PA-f-CNC_0.02 membrane upon 9 days of filtration (Figure 2.11). On the contrary, a control PA TFN membrane prepared with non-modified CNC (0.02 w/v% in water) (PA-CNC_0.02) exhibited no dramatic change in membrane performance (See Table 2.5 for detailed values), possibly due to the lack of CNC fillers [25,31,34]. Another control membrane, PA-PDPBA_0.02, prepared with PDPBA (0.02 w/v% in heptane), showed a dramatic reduction in salt rejection. Such a serious deterioration in salt rejection is presumably because monofunctional carboxylic acid consumes MPD or disturbs MPD diffusion to TMC/heptane solution during the interfacial reaction [53]. These results suggest that f-CNC is particularly efficient, likely because of the synergetic combination between hydrophilic CNC nano-structures and oleophilic surface functionalities. The extraordinary effects of f-CNC on membrane structures and/or separation performances are closely investigated in the later part of the manuscript.

2.3.3. Effect of f-CNC on the membrane morphology

Figure 2.12 shows cross-sectional TEM images of PA layer of the membranes. The neat PA membrane displayed relatively dense and thick PA layer, consistent with that of previously reported PA TFC membranes (Figure 2.12a) [25]. On the contrary, the PA layer became thinner and looser with the addition of f-CNC upon the IP process (Figure 2.12b-e). Specifically, the PA layer became an asymmetric structure, featuring a dense structure in the lower part and a loose structure in the upper part. The structurally loose upper layer containing many voids improves the permeability of the membrane [54-56]. The morphology distinguished by the overlap of leaf-like polyamides, featuring numerous cavities and frequently described as a belt-crater configuration, functions to increase the filtration area [37,38]. These voids play a vital role in diminishing the permeation resistance, thereby significantly enhancing the separation performance of the membrane [7,57]. When f-CNC loading is larger than a certain threshold (e.g., 0.03 w/v% in heptane), the PA layer became a little thicker. For example, the thickness of PA-f-CNC_0.05 membrane is larger than that of PA-f-CNC_0.03 membrane. This is possibly due to the large filler aggregation on the membrane surface and inside (Figure 2.9). The thickness of the control membrane, i.e., TFN membrane with 0.02 w/v%

non-modified CNC, was similar to that of neat PA membrane, indicating that the IP process is not significantly influenced by non-modified CNC (Figure 2.13). These results are consistent with cross-sectional SEM images (Figure 2.14), and these morphological changes suggests that the incorporation of f-CNC induced a thinner and asymmetric PA layer formation. This contributes to an increase in water permeability (up to 44%) by reducing water permeation resistance while retaining selectivity (only decreased up to maximum 0.1%) by thin and dense, lower part of PA layer [58].

One hypothesis for the morphological change is that well-dispersed f-CNC in heptane plays a critical role in formation of asymmetric PA layer upon IP process (Figure 2.1). In a typical IP process, because the solubility of diamines (*e.g.*, MPD) in organic solvent (*e.g.*, heptane) is larger than that of triacyl chloride (*e.g.*, TMC) in water, diamines diffuse to organic phase, leading to the polymerization at the interface yet at a little organic phase (Figure 2.1a) [59,60]. This typical diffusion-dependent polymerization dynamics could be modified by the interface adjustment: in this study, the f-CNC prefers to be located at the interface during the interfacial reaction process due to its amphiphilic nature, contributing to rerouting diffusion dynamics. We hypothesized two key roles of f-CNC at the interface: (i) it may lower the interfacial tension, promote MPD diffusion, and amplify the polymerization at the interface in early stage, resulting in the formation of a

thin and dense incipient PA layer that hinders further diamine diffusion and induces loose upper PA layer formation [54,61], and (ii) it forms a mesh-like network at the interface, which acts as a physical barrier confining the area for growth of PA layer, thereby reducing the PA layer thickness [34,58] (Figure 2.1b). These mechanics contribute the formation of thin asymmetric PA layer, exhibiting improved separation performances, and 0.02 w/v% could be the optimal concentration to achieve the lowest thickness [54].

To verify this hypothesis, the interfacial tension between water and heptane was evaluated by pendant drop method (Figures 2.15a and 2.16). The interfacial tension was dropped continuously upon increasing f-CNC loading in heptane, possibly indicating that f-CNC works like a surfactant [62]. The formation of the stable heptane droplets in water with f-CNC supports this result (Figure 2.17) [39]. Note that reduction in interfacial tension was observed when another powerful surfactant, sodium dodecyl sulfate (SDS), was dispersed in the aqueous phase simultaneously, indicating that f-CNC can also lower the interfacial tension in a real IP condition (Table 2.6). The diffusion rate of MPD into the organic phase depends on the interfacial resistance [58,60], and lowering interfacial tension generally promotes the diffusion of monomers [39,54,63]. In Figure 2.15b, we investigated the impact of f-CNC on MPD diffusion rate by monitoring the increase in UV absorbance. As MPD diffuses into the

organic phase, UV absorbance increases proportionally with the increasing MPD concentration [64,65]. Consequently, the slope of the absorption curve is positively correlated with the MPD diffusion rate [41,66]. We observed that as the concentration of f-CNC in the organic solvent layer increased, the slope of the absorption curve also increased. This finding indicates that the introduction of f-CNC led to a faster diffusion of MPD. Additionally, the increase in the degree of cross-linking of the membrane with f-CNC at the early stage of IP also indirectly supports the acceleration of MPD diffusion (Figure 2.18 and Table 2.7). The barrier effect of f-CNC could also affect the membrane surface morphologies (Figure 2.9): TFN membranes with f-CNC showed a larger and flatter leaf-like structure compared to a nodular structure observed in the TFC membrane, possibly suggesting that the f-CNC barrier restricts the vertical growth of PA during IP process [34,58]. The effect of f-CNC on the IP dynamics at the early stage was visualized by the support-free IP (Figures 2.19 and 2.20). The thickness of PA layer with f-CNC was smaller than that without f-CNC at all interfacial reaction period. This is consistent with high cross-linking density of TFN membranes with f-CNC (Figure 2.18 and Tables 2.7 and 2.8). These results suggest that f-CNC, located at or near the water/heptane interface, lowers the interfacial tension and acts as a physical barrier, manipulating the diffusion dynamics of monomers even from the early IP reaction stage. This leads to the formation

of a thin and asymmetric (i.e., lower dense and upper loose) PA layer. This structural feature significantly contributed to the improvement of water permeability. While it is possible to be contributed by f-CNC itself as other TFN membranes with CNC or CNC derivatives (Table 2.3), its contribution might be small given that the critical loading of f-CNC is extremely low.

The surface morphological changes are investigated by AFM analysis. Figure 2.21 shows the surface morphologies of membranes (Table 2.9 for detailed values). Surface roughness (R_q) increased with increasing f-CNC loadings: slight roughening at low f-CNC loadings while a significant roughening at high f-CNC loadings (i.e., 0.05 w/v%). The former small roughening is likely attributed to the surface roughening effects of introducing nanofillers. The f-CNC nanofiller itself forms a knoll that increases the roughness [16,31,34,67,68]. The latter large increase in roughness is due to the large-sized f-CNC agglomerates on the top surface that consistent with the surface SEM images (Figure 2.9). The control membrane, PA-CNC_0.02, showed larger increase in roughness than PA-f-CNC_0.02, possibly due to the nodular surface structure of PA-CNC_0.02 in contrast to the flatter leak-like surface structure of TFN membranes with f-CNC (Figures 2.9 and 2.13).

2.3.4. Antifouling and antibacterial properties

Figure 2.22 shows the normalized water permeability of the membranes with a feed solution containing a model foulant (1000 ppm BSA) (Table 2.10 for detailed values). The flux decline ratio (*FDR*) and flux recovery ratio (*FRR*) were evaluated after three cycles of BSA fouling and washing. The PA-f-CNC_0.02 membrane showed lower *FDR* and higher *FRR* compared to the neat PA membrane, indicating that f-CNC imparts antifouling properties, i.e., less foulant adhesion and easier foulant removal [68,69]. In general, adhesion of foulant on a membrane surface is governed by surface hydrophilicity and surface energy [11]. More hydrophilic surface has better antifouling property given that the surface favors interaction with water molecule rather than a foulant. The TFN membranes containing f-CNC exhibited contact angle values slightly lower than the neat PA membrane, indicating that incorporation of f-CNC slightly increases the hydrophilicity of the membranes (Figure 2.23). It is likely attributed to the combination of partial hydrophilicity of f-CNC and slightly increased roughness [70]. This is consistent with surface energy evaluation (Table 2.11). The surface energy of the PA-f-CNC_0.02 membrane is closer to that of water (72 mN/m), suggesting that a hydration layer is generated more effectively, thereby preventing adhesion of foulants [71,72]. Another control PA membrane with non-modified CNC (PA-CNC_0.02) also showed the better antifouling performance due to the hydrophilic nature of CNC (Figure

2.24) [23,31]. However, the absolute water flux of the TFN membrane with f-CNC after 24 h of filtration is larger than that of the control membrane with CNC given the higher initial water permeability, suggesting that the TFN membrane with f-CNC is likely more useful under the practical condition.

E. coli was selected as model bacteria to evaluate the antibacterial properties of the membranes. Figure 2.25a shows the antibacterial property of the membranes by measuring the death rate of the bacteria after 24 h culturing on the membrane surface. The antibacterial property is governed by adhesion of bacteria on a surface and bacterial mortality. No bacterial mortality was observed after culturing on the neat PA membrane surface. In contrast, all TFN membranes, with CNC and f-CNC, exhibited increased antibacterial properties, suggesting the presence of inherent biocidal and low bio-adhesion properties. This is possibly attributed to the intrinsic biocidal activity of the CNC with rod-like structure, comparable to CNT and GO: cell membranes of the bacteria could be punctured by the sharp and stiff materials [24,28]. The TFN membranes with f-CNC showed higher mortality than that with CNC at the same loading (0.02 w/v%). This indicates long alkyl chain, i.e., PDP moiety, on the f-CNC provides an additional antibacterial property. It has been known that long saturated alkyl chains make cell punctuation easier due to the similar chemical nature with

lipid bilayer of the cell membrane [28,73,74]. The synergistic combination of rod-shaped CNC and PDP with long alkyl chains facilitated the disruption of bacterial membranes, thereby increasing the antibacterial properties. Given that the antibacterial activity is determined by the complicated combinations of a various factors including surface hydrophilicity as well as surface morphology, there is a certain fluctuation in antibacterial properties at different f-CNC loadings. Typically, the antimicrobial activity is improved by achieving a higher dispersion of nanofillers or by obtaining a smoother and more hydrophilic membrane surface that reduces bacterial adhesion [75,76]. The antibacterial properties were visualized by CLSM analysis (Figure 2.25b). A number of green spots (live *E. coli*) with no red spot (dead *E. coli*) were observed on the neat PA membrane surface while small number of green and red spots were observed on the TFN membranes with f-CNC. Reduction in the number of spots (i.e., number of attached bacteria) is due to the mitigated bacterial adhesion, and the red spots clearly indicate the biocidal properties of the membrane surface. Furthermore, in the antibiofouling test with a bacterial feed solution, PA-f-CNC_0.02 showed a lower *FDR* and higher *FRR* values compared to neat PA, indicating that f-CNC could inhibit the attachment of microorganism and the formation of biofilm on the membrane surface even under the practical desalination process (Figure 2.26).

2.4. Conclusions

This study demonstrates that asymmetric PA layer with high permeability and salt rejection is fabricated by simply incorporating amphiphilic f-CNC in the organic phase upon the established and practical IP process. The f-CNC near the interface manipulates the diffusion of the monomer upon IP: (i) it lowers the interfacial tension and facilitates the diffusion, forming a dense initial PA layer, and (ii) it restricts the vertical expansion of the PA layer in the upper part, reducing the thickness of the selective layer. This enables to generate thin asymmetric PA layer that exhibits improved water permeability (44 % increase) while maintaining high NaCl salt rejection (98.6 %). Furthermore, the TFN membrane with f-CNC showed enhanced antifouling and antibacterial properties, originated from specific nature of f-CNC, suggesting that the TFN membrane is robust under the practical filtration process. The materials and approaches described in this study provide insights into the utilization of nanofillers for engineering diffusion-based IP process, diverged from the conventional nanofiller applications. We believe this study highlights the potentially scalable, efficient, and eco-friendly fabrication of advanced PA membranes having antibacterial property that improve production of fresh water.

2.5. References

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Table 2.1. Atomic compositions from XPS analysis of CNC and f-CNC.

Samples	Atomic compositions (At.%)	
	C 1s	O 1s
CNC	58.96	41.04
f-CNC	76.09	23.91

Table 2.2. The average apparent particle size from dynamic light scattering of CNC and f-CNC.

Sample	Particle size (nm)
CNC in water	83.7 ± 18.7
f-CNC in heptane	87.0 ± 11.0

Table 2.3. Separation performance of neat PA and PA with CNC materials membranes reported in previous studies.

Optimum CNC Loading	TFC (neat PA)		TFN (PA with CNC materials)		Ref.
	Water Flux ($\text{Lm}^{-2}\text{h}^{-1}\text{bar}^{-1}$)	Salt Rejection (%)	Water Flux ($\text{Lm}^{-2}\text{h}^{-1}\text{bar}^{-1}$)	Salt Rejection (%)	
CNC (0.10 w/v%)	1.50	98.50	3.15	97.80	[R1]
CNC (0.10 wt%)	0.24	97.53	0.42	94.40	[R2]
TEMPO-oxidized CNC (0.50 wt%)	0.24	97.53	0.63	98.98	[R2]
Sawdust derived CNC (0.21 wt%)	0.83	99.00	1.44	98.30	[R3]
CNC (0.1 wt%)	1.27	98.00	1.80	97.10	[R4]
Acetylated CNC2 (0.4 wt%)	1.27	98.00	2.60	98.00	[R4]
Acetylated CNC4 (0.4 wt%)	1.27	98.00	1.90	98.70	[R4]
CysCNC (0.2 wt%)	6.00	~ 43.00	15.7	43.3	[R5]
f-CNC (0.02 w/v%)	1.26	98.65	1.82	98.60	This work

[R1] Desalination 439 (2018) 179-187

[R2] Nanomaterials 9 (2019) 125

[R3] Alex. Eng. J. 59 (2020) 4201-4210

[R4] Desalination 538 (2022) 115900

[R5] J. Membr. Sci. 670 (2023) 121369

Table 2.4. The water (A) and salt (B) permeability coefficient of neat PA and PA-f-CNC_0.02 membranes.

Samples	Water permeability coefficient, A (L m ⁻² h ⁻¹ bar ⁻¹)	Salt permeability coefficient, B (L m ⁻² h ⁻¹)	A/B (bar ⁻¹)
neat PA	1.26	0.194	6.49
PA-f-CNC_0.02	1.82	0.265	6.87

Table 2.5. Pure water flux and salt rejection of PA-CNC_0.02 and PA-PDPBA_0.02 membranes (under lab-scale cross-flow system, pressure: 15.5 bar, flow rate: 0.8 L/min, and feed solution: 2000 ppm NaCl solution).

Samples	Water flux (L m ⁻² h ⁻¹ bar ⁻¹)	Salt rejection (%)
PA-CNC_0.02	1.28 ± 0.04	99.1 ± 0.11
PA-PDPBA_0.02	5.99 ± 1.92	20.8 ± 9.76

Table 2.6. Interfacial tension of water/heptane with CNC, f-CNC and SDS.

Water phase (w/v%)	Heptane phase (w/v%)	Interfacial tension (mN/m)
-	-	47.4 ± 0.41
-	f-CNC 0.01	38.1 ± 0.26
-	f-CNC 0.02	36.0 ± 0.26
-	f-CNC 0.03	33.4 ± 0.17
-	f-CNC 0.05	31.0 ± 0.15
CNC 0.02	-	48.1 ± 0.25
SDS 0.1	-	6.1 ± 0.15
SDS 0.1	f-CNC 0.02	3.8 ± 0.28

Table 2.7. The surface atomic compositions from XPS analysis of neat PA membranes and PA-f-CNC membranes fabricated through interfacial polymerization for 5 s, 10 s, 20 s, and 30 s.

Samples	Atomic compositions (At.%)			
	C 1s	O 1s	N 1s	O/N ratio
PA_5s	74.03	18.93	7.04	2.69
PA_10s	72.96	19.15	7.88	2.43
PA_20s	74.47	16.74	8.78	1.91
PA_30s	74.39	16.61	8.99	1.85
PA-f-CNC_0.02_5s	73.13	18.02	8.85	2.04
PA-f-CNC_0.02_10s	73.03	17.86	9.10	1.96
PA-f-CNC_0.02_20s	74.10	16.47	9.42	1.75
PA-f-CNC_0.02_30s	74.06	16.13	9.80	1.65
PA-f-CNC_0.03_5s	73.33	18.33	8.34	2.20
PA-f-CNC_0.03_10s	73.40	18.31	8.29	2.21
PA-f-CNC_0.03_20s	73.38	17.02	9.60	1.77
PA-f-CNC_0.03_30s	74.18	16.22	9.60	1.69

Table 2.8. The surface atomic compositions from XPS analysis of neat PA and PA-f-CNC membranes.

Samples	Atomic compositions (At.%)			
	C 1s	O 1s	N 1s	O/N ratio
PA	71.4	18.49	10.11	1.83
PA-f-CNC_0.01	71.28	18.44	10.28	1.79
PA-f-CNC_0.02	72.74	16.65	10.61	1.57
PA-f-CNC_0.03	72.38	16.94	10.68	1.59
PA-f-CNC_0.05	73.12	16.45	10.43	1.58

Table 2.9. Surface roughness values of TFC and TFN membranes.

Samples	R_q^a (nm)	R_{pv}^b (nm)
PA	74.761	603.746
PA-f-CNC_0.01	77.844	639.568
PA-f-CNC_0.02	82.286	628.368
PA-f-CNC_0.03	94.255	688.305
PA-f-CNC_0.05	110.279	885.887
PA-CNC_0.02	110.133	789.943

^a R_q : Root mean squared roughness

^b R_{pv} : difference between maximum peak height and maximum valley depth

Table 2.10. The values of *FDR* and *FRR* of neat PA, PA-f-CNC_0.02, and PA-CNC_0.02 membranes for antifouling test with BSA feed solution.

Samples	<i>FDR</i> (%)			<i>FRR</i> (%) – Cycle 3
	Cycle 1	Cycle 2	Cycle 3	
PA	19.6	19.7	23.1	79.2
PA-f-CNC_0.02	12.4	13.4	18.7	83.7
PA-CNC_0.02	7.1	8.2	12.8	91.9

Table 2.11. Contact angle and surface energy values of TFC and TFN membranes.

Samples	Contact angle (°)		Surface energy (mN/m)		
	Water	Diiodomethane	Total	Polar	Dispersive
PA	43.69 ± 0.06	45.2 ± 1.86	60.05 ± 1.52	36.91 ± 1.00	23.14 ± 0.52
PA -f-CNC_0.01	39.28 ± 0.55	37.64 ± 1.75	64.47 ± 1.41	40.78 ± 0.85	23.69 ± 0.56
PA -f-CNC_0.02	36.38 ± 0.14	23.55 ± 1.01	69.07 ± 0.52	46.66 ± 0.34	22.42 ± 0.18
PA -f-CNC_0.03	39.96 ± 0.11	43.88 ± 1.16	62.55 ± 0.95	37.61 ± 0.62	24.94 ± 0.34
PA -f-CNC_0.05	42.30 ± 0.44	48.60 ± 0.38	60.01 ± 0.56	35.05 ± 0.21	24.96 ± 0.35
PA -CNC_0.02	30.17 ± 0.76	28.65 ± 1.53	71.08 ± 1.14	44.77 ± 0.61	26.31 ± 0.53

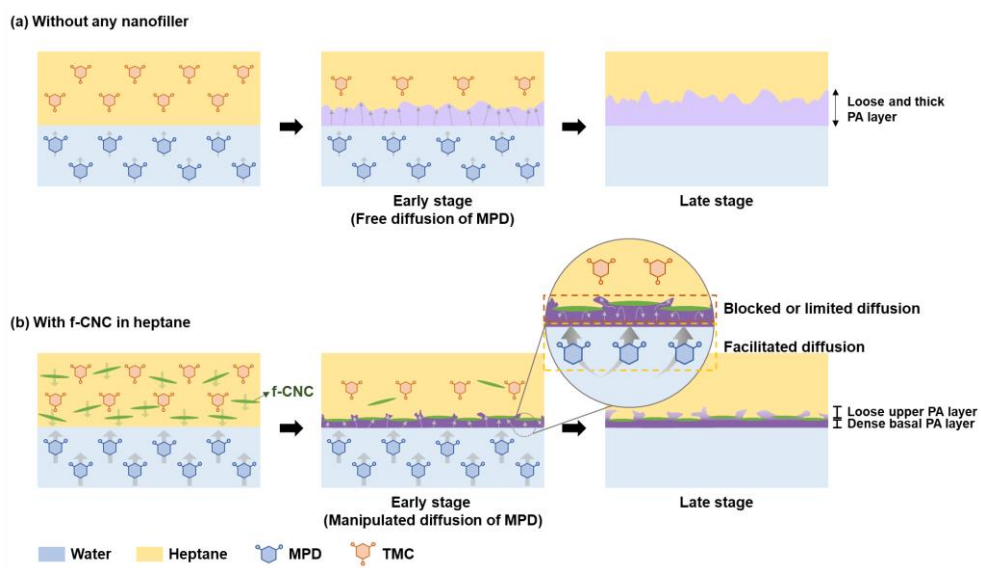


Figure 2.1. Schematic illustration of the interfacial polymerization. (a) without and (b) with f-CNC in heptane phase.

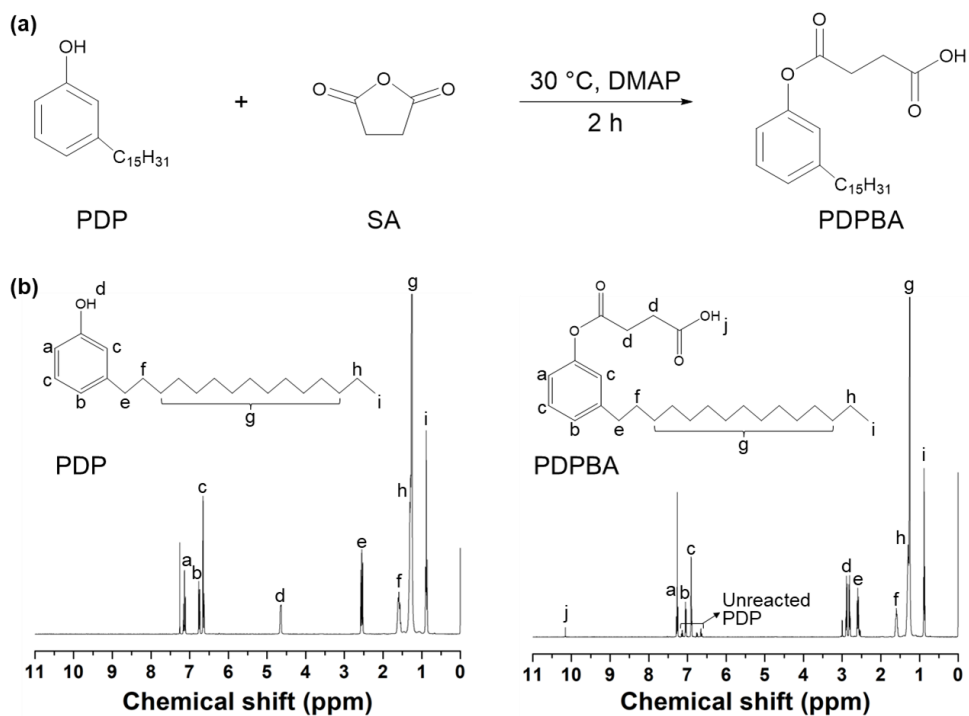
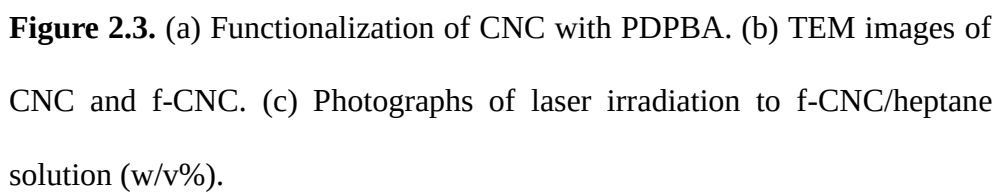


Figure 2.2. (a) Synthesis of PDPBA. (b) ^1H -NMR spectra of PDP and PDPBA.



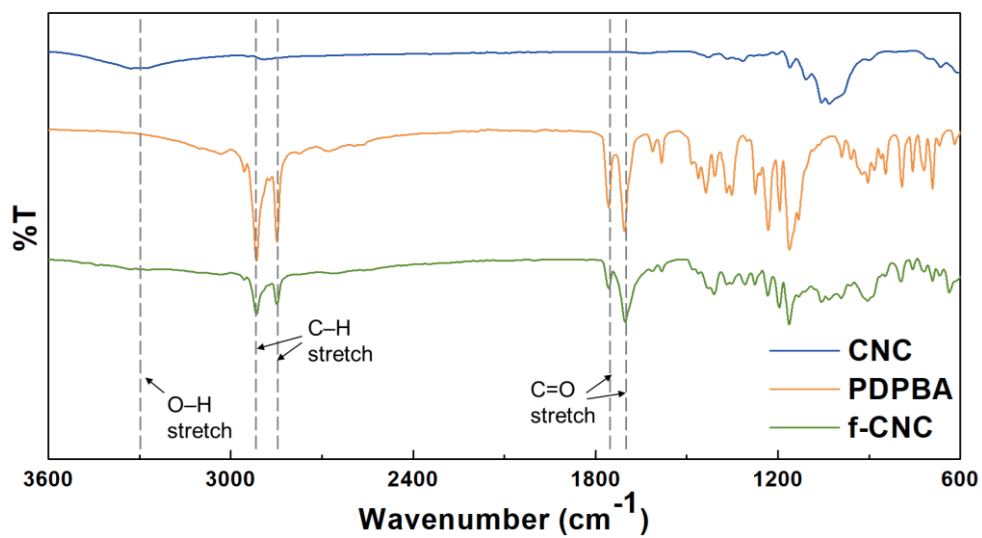


Figure 2.4. FT-IR spectra of CNC, PDPBA, and f-CNC.

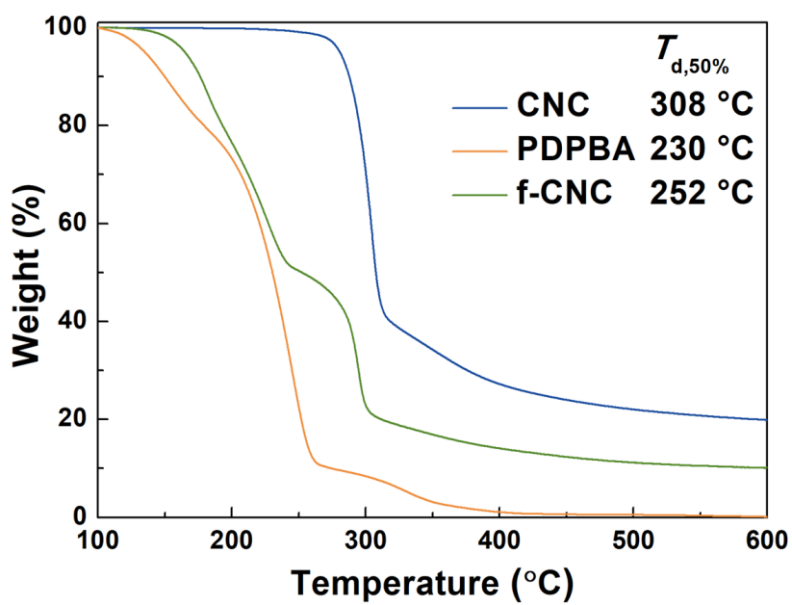


Figure 2.5. TGA curves of CNC, PDPBA, and f-CNC.

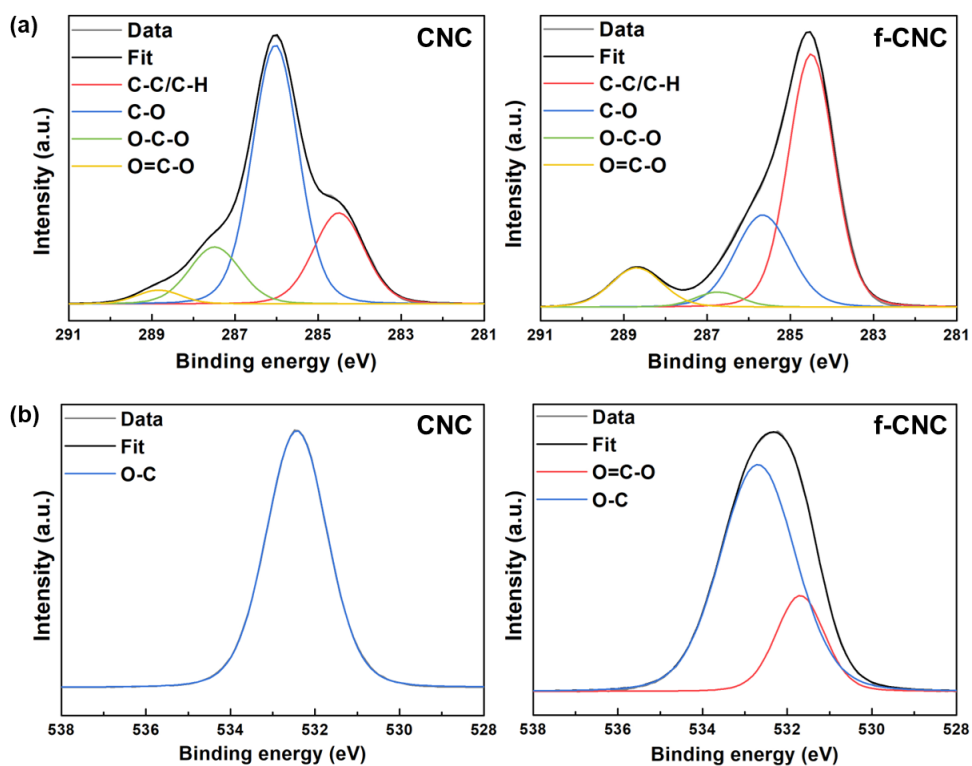


Figure 2.6. XPS analysis of CNC and f-CNC. (a) C 1s and (b) O 1s spectra.

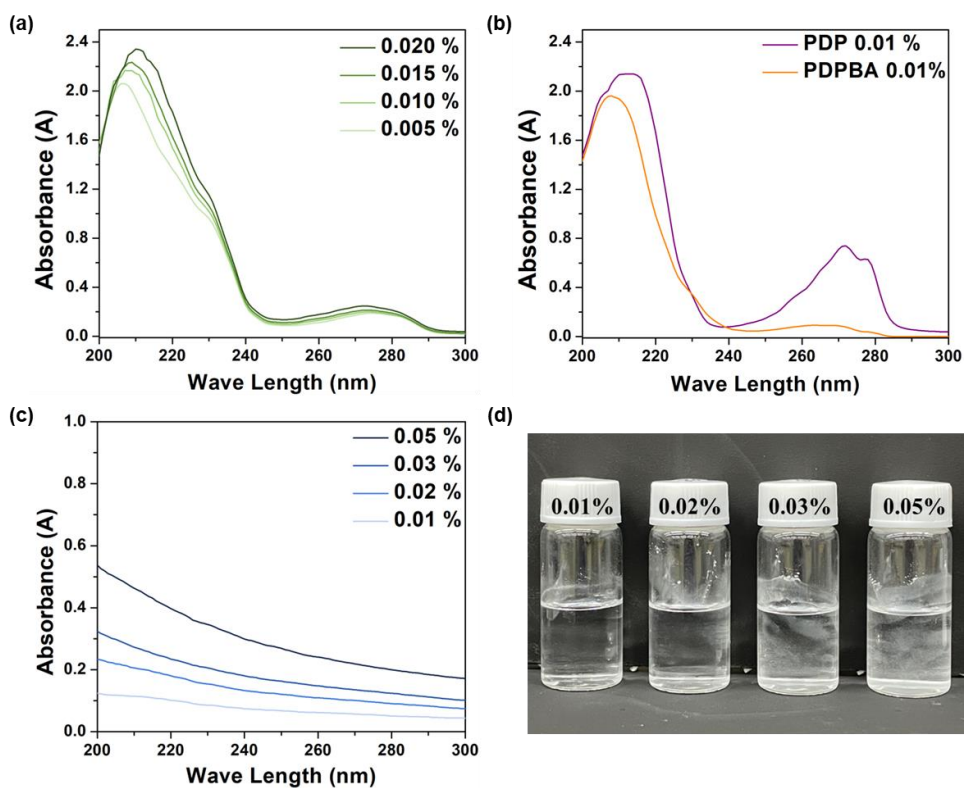


Figure 2.7. UV-Vis absorption spectra of (a) f-CNC dispersed in heptane, (b) PDP and PDPBA dispersed in heptane, and (c) CNC dispersed in water. (d) Photographs of CNC/heptane solution. All concentration unit is w/v%.

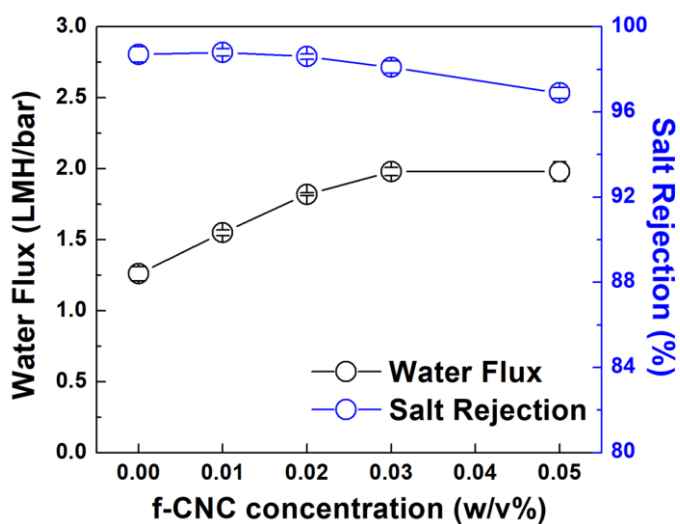


Figure 2.8. Pure water flux and salt rejection of TFN with f-CNC membranes (under lab-scale cross-flow system, pressure: 15.5 bar, flow rate: 0.8 L/min, and feed solution: 2000 ppm NaCl solution).

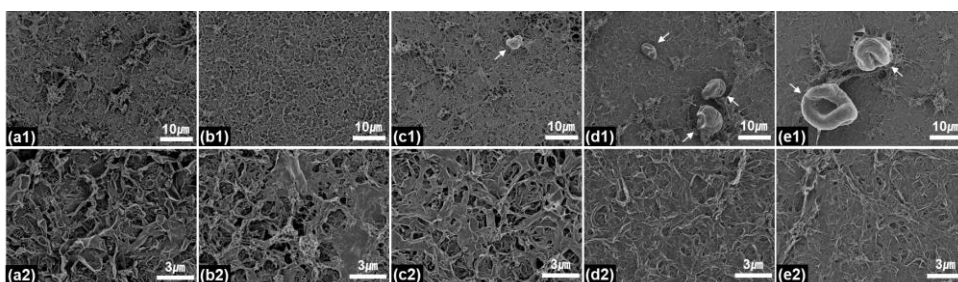


Figure 2.9. Surface SEM images of (a1-a2) neat PA, (b1-b2) PA-f-CNC_0.01, (c1-c2) PA-f-CNC_0.02, (d1-d2) PA-f-CNC_0.03, and (e1-e2) PA-f-CNC_0.05 membranes. White arrows indicate aggregated f-CNC particles.

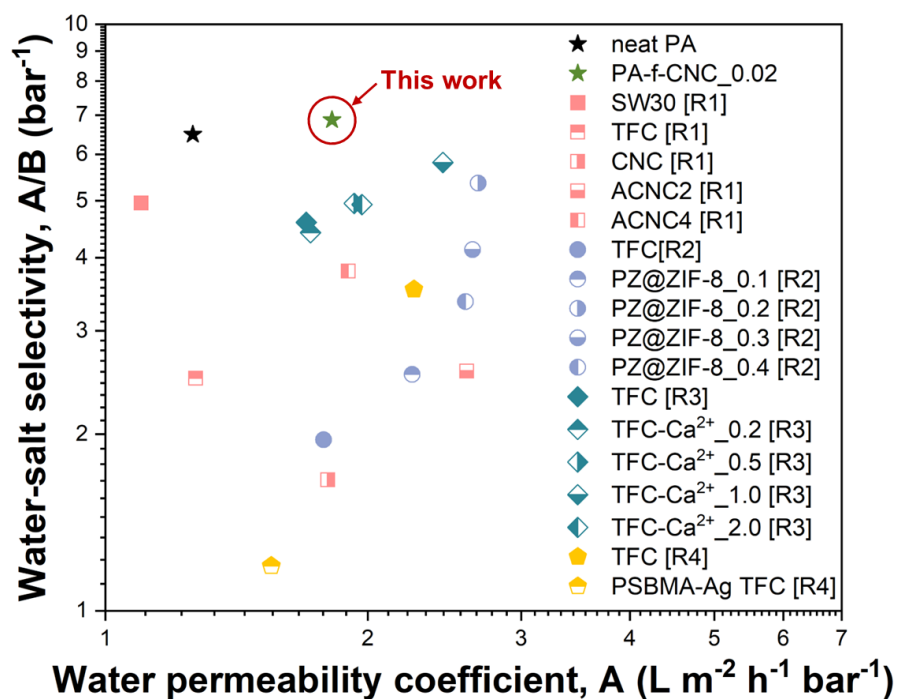


Figure 2.10. Water permeability coefficient (A) and water-salt selectivity (A/B) of TFC and TFN membranes [R1-R4].

[R1] Desalination 538 (2022) 115900.

[R2] Environ. Sci. Technol. 55 (2021) 5324-5334.

[R3] Environ. Sci. Technol. 53 (2019) 4371-4379.

[R4] Environ. Sci. Technol. 51 (2017) 182-191.

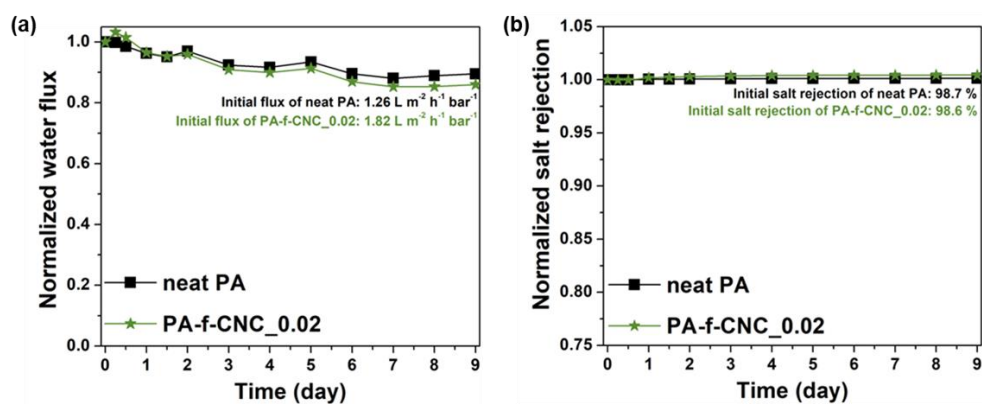


Figure 2.11. Normalized (a) water flux and (b) salt rejection of neat PA and PA-f-CNC_0.02 membranes tested with 2000 ppm NaCl aqueous solution for 9 days (evaluated by lab-scale cross-flow system, pressure: 15.5 bar, and flow rate: 0.8 L/min).

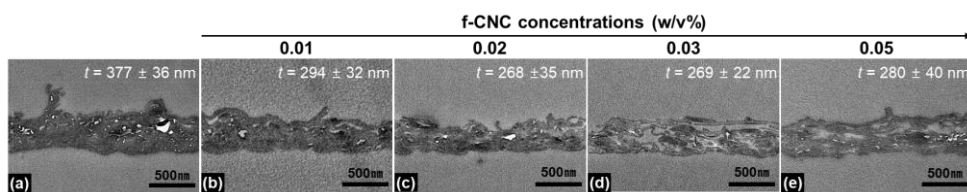


Figure 2.12. Cross-sectional TEM images of PA layer of the membranes with different f-CNC concentrations: (a) neat PA, (b) PA-f-CNC_0.01, (c) PA-f-CNC_0.02, (d) PA-f-CNC_0.03, and (e) PA-f-CNC_0.05 membranes.

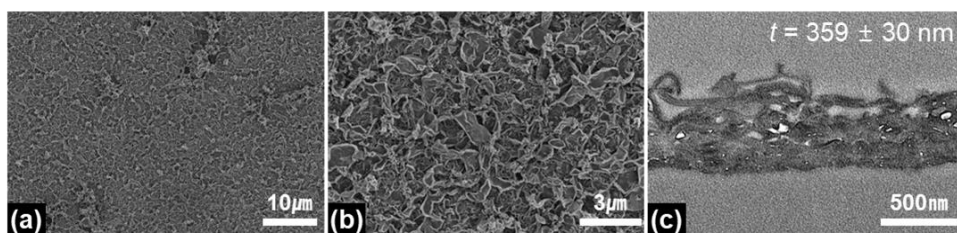


Figure 2.13. (a-b) Surface SEM images and (c) cross-sectional TEM images of PA-CNC_0.02 membrane.

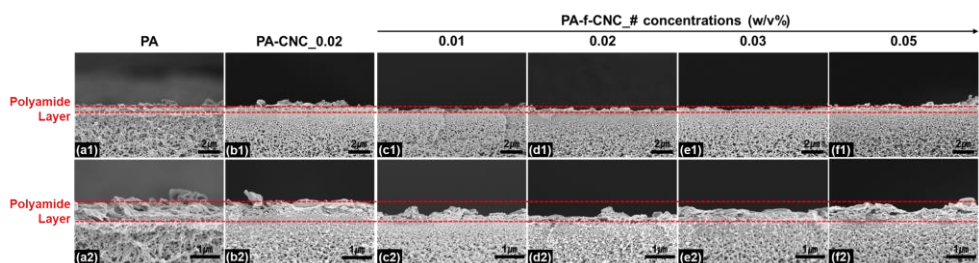


Figure 2.14. Cross-sectional SEM images of (a1-a2) neat PA, (b1-b2) PA-CNC_0.02, (c1-c2) PA-f-CNC_0.01, (d1-d2) PA-f-CNC_0.02, (e1-e2) PA-f-CNC_0.03, and (f1-f2) PA-f-CNC_0.05 membranes.

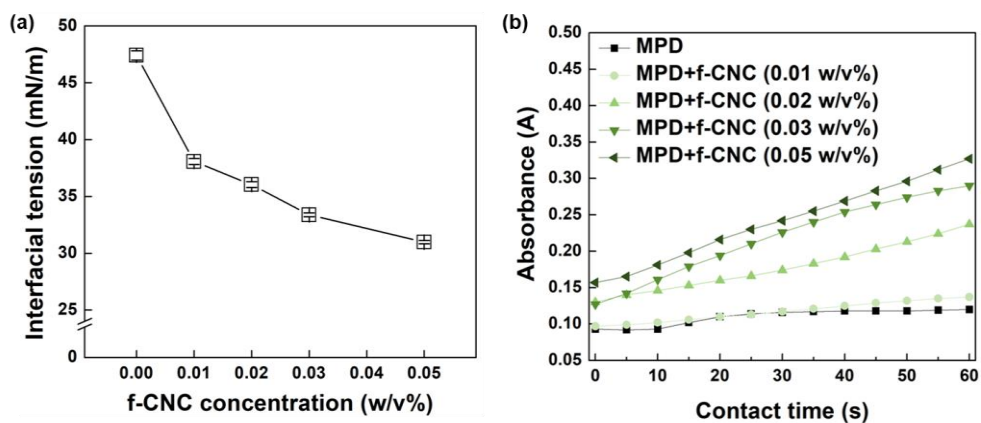


Figure 2.15. (a) Interfacial tension of water/heptane with f-CNC. (b) Effect of f-CNC on the diffusion rate of MPD.

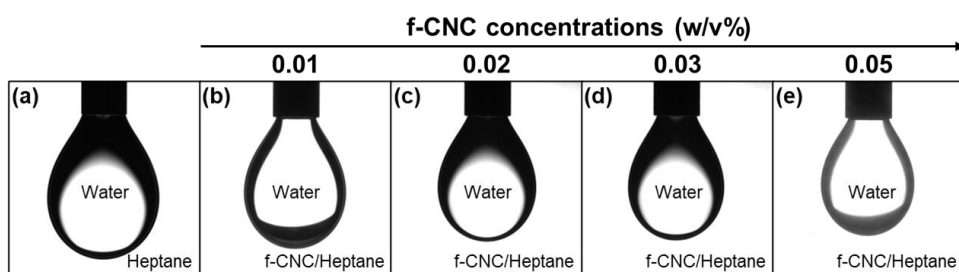


Figure 2.16. Images of water droplet in (a) heptane and (b-e) f-CNC/heptane solutions (w/v%).

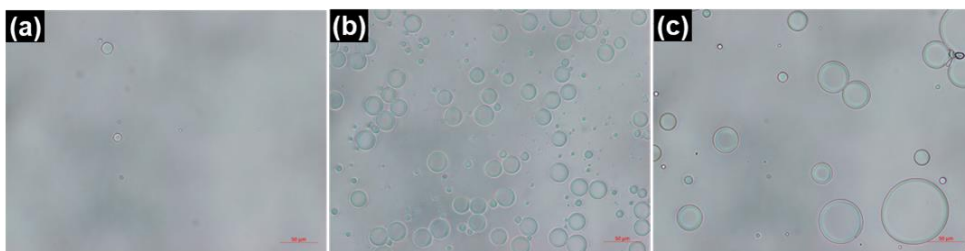


Figure 2.17. Optical microscope images of heptane droplets in water (a) without any additives, (b) with 1 mg/mL of SDS in water, and (c) with 1 mg/mL of f-CNC in heptane.

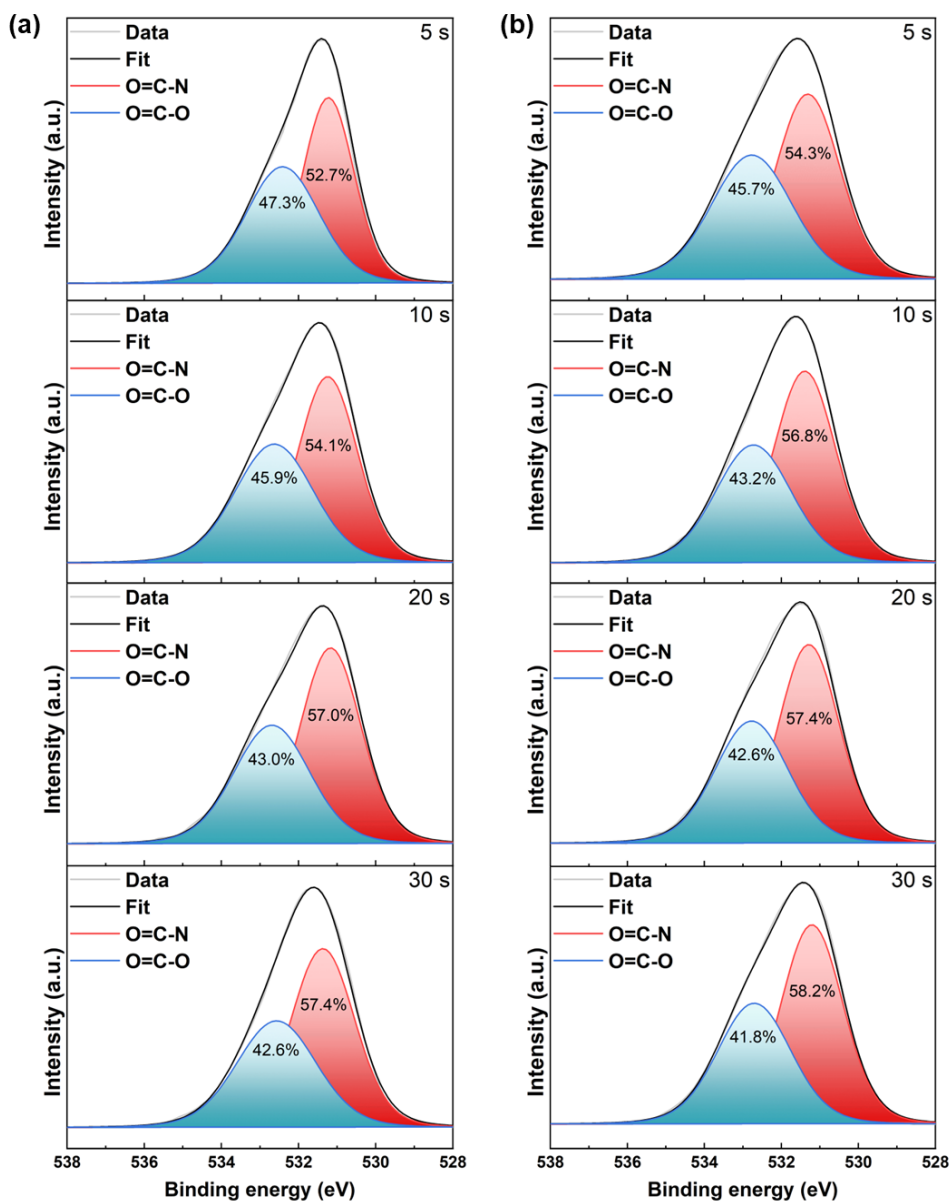


Figure 2.18. XPS O 1s spectra of (a) neat PA membranes and (b) PA-f-CNC_{0.03} membranes at the early stage of interfacial polymerization.

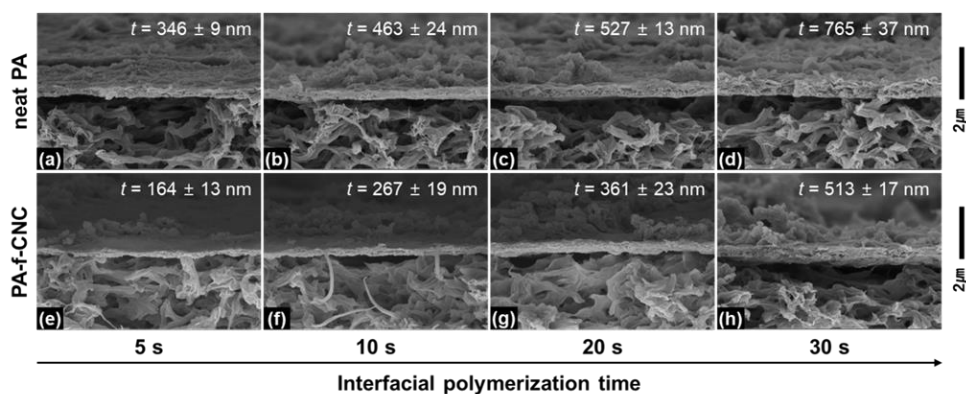


Figure 2.19. Cross-sectional SEM images of (a-d) neat PA membranes and (e-h) PA-f-CNC_0.03 membranes at the early stage of interfacial polymerization (PA layers were synthesized through support-free IP and transferred onto a nylon substrate for observation).

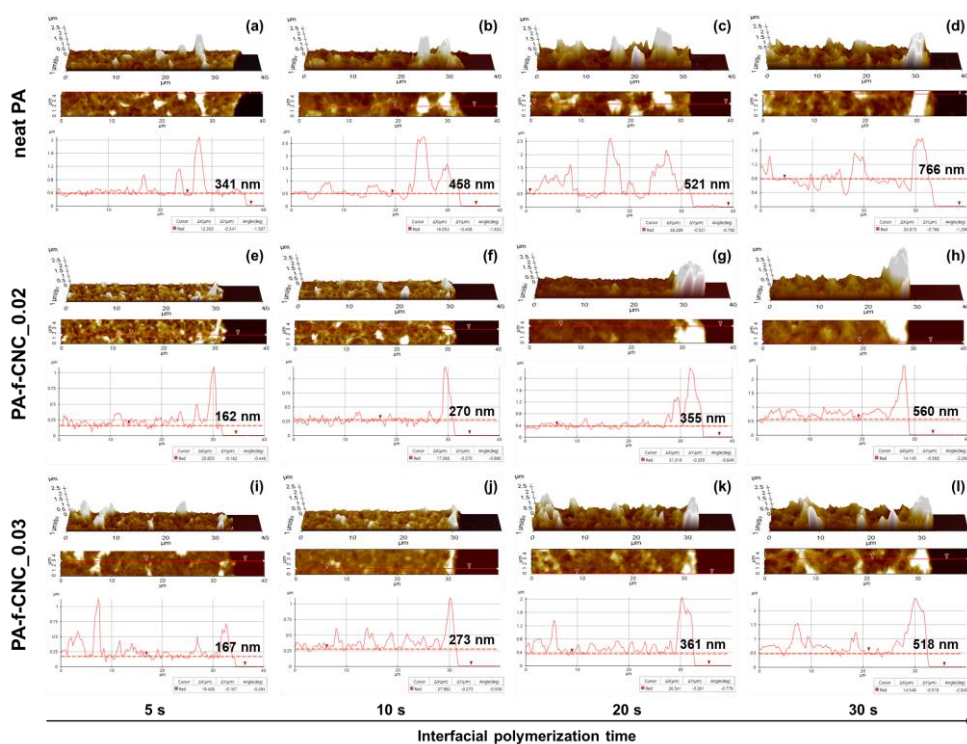


Figure 2.20. AFM 3D images and height profiles of (a-d) neat PA membranes, (e-h) PA-f-CNC_0.02, and (i-l) PA-f-CNC_0.03 membranes at the early stage of interfacial polymerization.

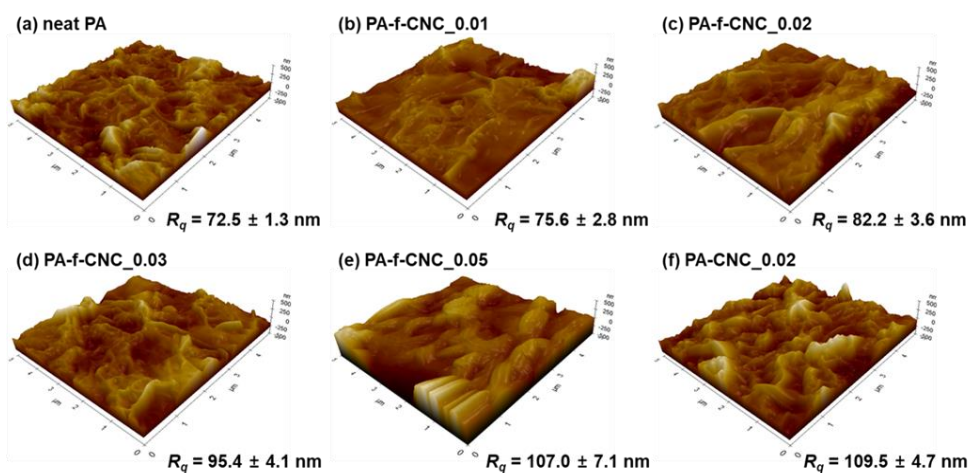


Figure 2.21. AFM images of (a) neat PA, (b) PA-f-CNC_0.01, (c) PA-f-CNC_0.02, (d) PA-f-CNC_0.03, (e) PA-f-CNC_0.05, and (f) PA-CNC_0.02 membranes.

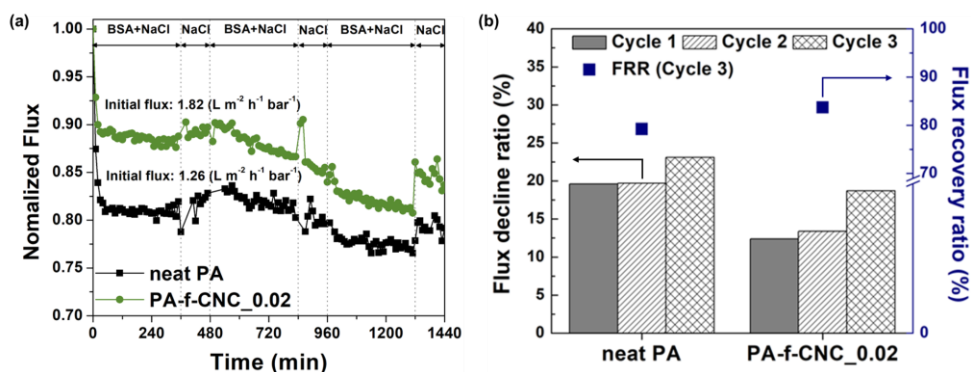


Figure 2.22. (a) Normalized water flux of neat PA and PA-f-CNC_0.02 membranes tested with 1000 ppm BSA and 2000 ppm NaCl aqueous solution (evaluated by lab-scale cross-flow system, pressure: 15.5 bar, and flow rate: 0.8 L/min). (b) The values of *FDR* and *FRR* of neat PA and PA-f-CNC_0.02 membranes.

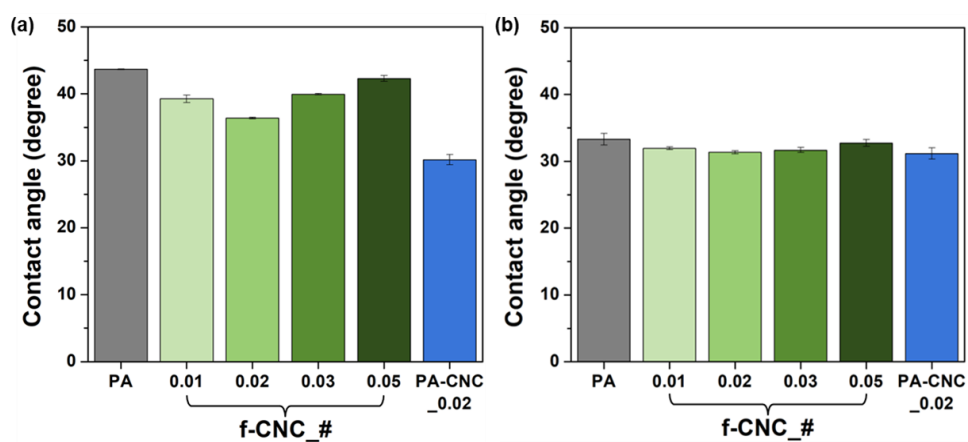


Figure 2.23. Surface water contact angle of TFC and TFN membranes measured by (a) the sessile drop method and (b) the captive bubble method.

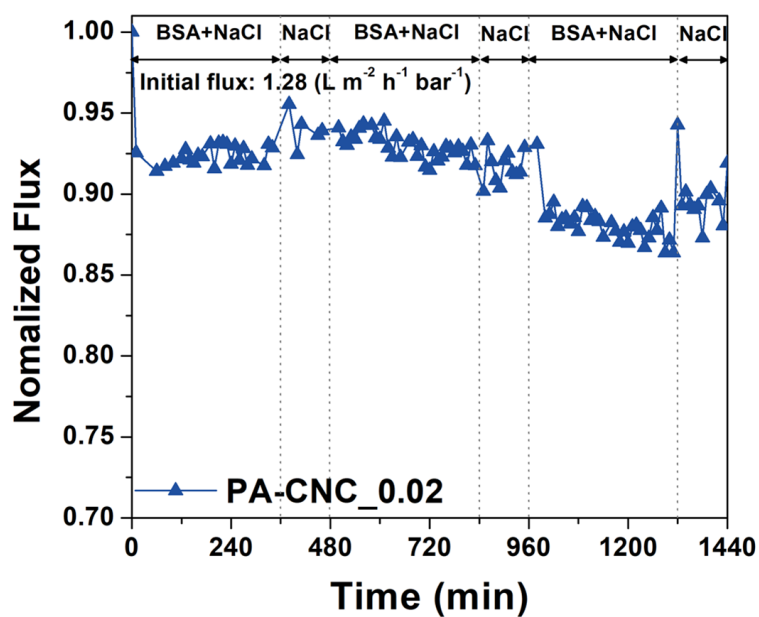


Figure 2.24. Normalized water flux of PA-CNC_0.02 membrane tested with 1000 ppm BSA and 2000 ppm NaCl aqueous solution (evaluated by lab-scale cross-flow system, pressure: 15.5 bar, and flow rate: 0.8 L/min).

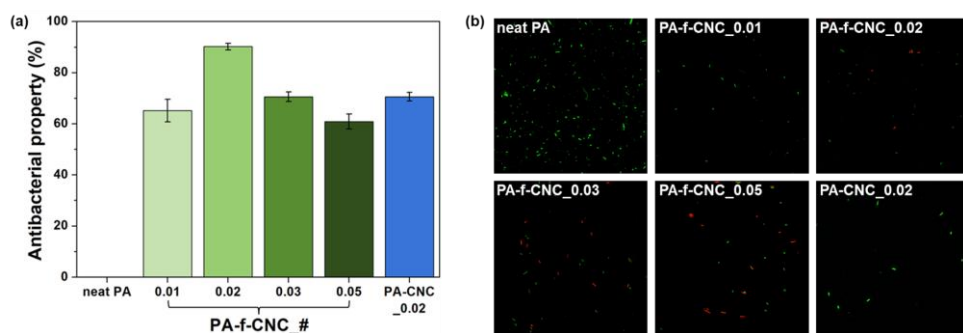


Figure 2.25. (a) Antibacterial property on the surface of TFC and TFN membranes against *E. coli*. (b) CLSM images of TFC and TFN membranes after static bacteria adhesion test using *E. coli*.

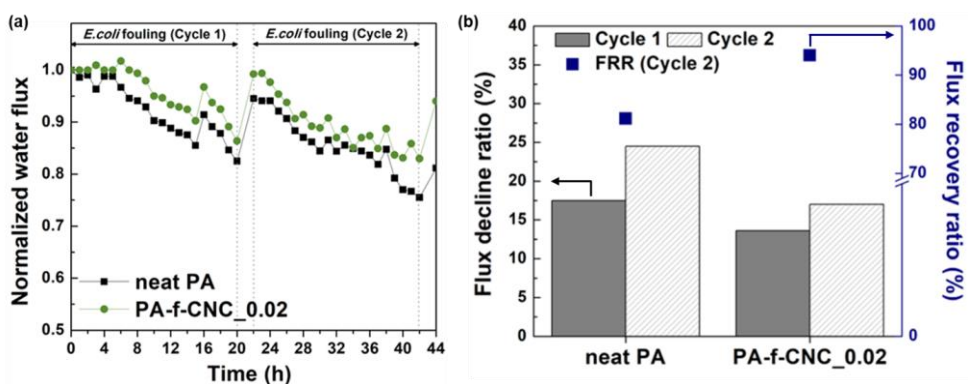


Figure 2.26. Normalized water flux of neat PA and PA-f-CNC_0.02 membranes tested with 10^6 CFU/ml of *E. coli* in PBS solution (evaluated by lab-scale cross-flow system, pressure: 15.5 bar, and flow rate: 0.4 L/min). (b) The values of *FDR* and *FRR* of neat PA and PA-f-CNC_0.02 membranes.

Chapter 3

**Highly dispersive surface-modified zinc silicate glass
powders manifesting dual mechanism antimicrobial
activity in PP composites**

3.1. Introduction

Polypropylene (PP) plays a crucial role in various facets of modern life, spanning from packaging, household appliances, military, to even advanced electronic devices. As the second most produced plastic after polyethylene, PP boasts global market volume of 80.7 million metric tons (MMT) in 2023, representing 20 % of global plastic consumption [1, 2]. Its simple chemical structure, characterized by a saturated hydrocarbon with a methyl side chain, enables remarkably inexpensive and versatile applicability. However, the absence of inherent functionality in its chemical structure renders PP susceptible to microbial contamination, primarily due to the lack of antibacterial activity. Additionally, its intrinsic hydrophobic nature fosters microbial adhesion on its surface, leading to rampant microbial proliferation.

Various strategies have been employed to address the susceptibility of PP towards microbial contamination, including incorporating antimicrobial organic/inorganic materials, grafting antimicrobial components, and modulating the surface properties, thereby expanding its utility and lifespan [3-5]. Among these strategies, the incorporation of inorganic antimicrobial fillers emerges as an efficient approach with feasible preparation. Therefore, a wide range of inorganic materials such as zinc oxide, silver, and copper have been utilized as antimicrobial fillers for PP, featuring strong

antimicrobial activity through the release of metal ions [6-8]. In our previous research, a zinc silicate glass powder (ZS) was employed as an antimicrobial filler for PP, demonstrating its superior efficiency through comparative analysis with commercial zinc oxide [9]. Its superior antimicrobial activity was attributed to the release of zinc ions, combined with its hydrophilic and negatively charged surface which effectively inhibited bacterial adhesion.

Despite their advantageous features, inorganic antimicrobial fillers intrinsically face a notable limitation. Since antimicrobial activity of these fillers relies on the release of metal ions, it inevitably diminishes over time due to the continuous depletion of metal ions, thereby restricting their long-term effectiveness. To address this issue, extensive efforts have been devoted to improving their long-term efficiency by controlling the release of metal ions and introducing additional antimicrobial components [10, 11]. Specifically, surface modification of inorganic fillers with organic antimicrobial compounds facilitates synergistic collaboration between two mechanisms—contact killing by the organic antimicrobial components and antimicrobial ion release from the inorganic fillers—ensuring persistent antimicrobial activity [12]. Furthermore, the surface modification of inorganic fillers using organic compounds significantly alters their surface properties, affecting both filler-filler and filler-polymer interactions [13].

Therefore, the systematic modulation of filler surface properties through surface modification can promote more uniform dispersion of fillers within polymer matrix, ultimately optimizing the overall performance of polymer composites.

To achieve persistent antimicrobial activity and uniform filler dispersion simultaneously, we employed silanization as a surface modification method, utilizing a ZS as an antimicrobial inorganic filler. As shown in Figure 3.1, we designed a series of silane coupling agents denoted as #ACs, wherein # represents the carbon number of alkyl chain attached to the amine. All #ACs feature a quaternary ammonium group as the antimicrobial component, with alkyl chains of varying lengths, which enables the systematic modulation of filler surface properties. We believe that #ACs offers a valuable model system for investigating how surface properties of fillers influence overall performances of polymer composites, especially with respect to their antimicrobial activity.

3.2. Experimental

3.2.1. Materials

Polypropylene (HiPrene ® M560) was supplied by GS Caltex. Dimethyloctadecyl[3-(trimethoxysilyl)propyl]ammonium chloride solution (42 wt.% in methanol), (3-chloropropyl)trimethoxysilane, *N,N*-dimethylbutylamine, *N,N*-dimethylhexylamine, *N,N*-dimethyloctylamine, phosphate buffer saline (PBS, pH 7.4), D-(+)-glucose, calcium chloride (CaCl₂), *p*-xylene (99 %), and sodium chloride (NaCl) were purchased from Aldrich and used as received. Trimethyl[3-(trimethoxysilyl)propyl]ammonium chloride (50 % in methanol) was purchased from TCI and used as received. *Escherichia coli* (*E. coli*; ATCC 700926) was obtained from American Type Culture Collection (ATCC). Live/Dead Viability/Cytotoxicity kit for mammalian cells and tryptone soya broth (Soyabean casein digest medium, SCDLP) were purchased from Thermo Fisher Scientific. Difco LB agar and Difco nutrient broth were obtained from Becton, Dickinson and Company (BD). All other chemicals and solvents were purchased from standard vendors and used as received.

3.2.2. Preparation of zinc silicate glass powder (ZS)

Zinc silicate glass powder was prepared through the melt-quenching

process, employing a procedure previously reported with a modification in the separation process [9]. Briefly, a 60.0 g batch of precursor materials, including ZnO, SiO₂, K₂CO₃, CaCO₃, Na₂CO₃, and B₂O₃, was mixed using a 3D mixer for 5 h to obtain a homogeneous mixture, with specific compositions listed in Table 3.1. After transferring the mixture into a Pt crucible, it was placed in an electric furnace, heated to 1300 °C at a heating rate of 10 °C min⁻¹, and maintained for 1 h. The molten glass was then poured into a pre-heated graphite mold, annealed at 550 °C for 1 h, and cooled slowly to room temperature to remove thermal residual stress. The resultant was crushed by ball milling to obtain glass powder. It is noted that the obtained glass powder exhibits a larger average diameter value (1.47 μm) than that of a glass powder (0.34 μm) utilized in our previous study due to the omission of the separation procedure using an air classifier.

3.2.3. Synthesis of dimethylalkyl[3-(trimethoxysilyl)propyl]ammonium chloride (#AC)

Dimethylalkyl[3-(trimethoxysilyl)propyl]ammonium chloride compounds with three distinct alkyl groups—specifically butyl, hexyl, and octyl—were synthesized and denoted as #AC, where the number # represents the carbon number of the respective alkyl group. Herein, the synthetic procedure for 4AC is described as a representative example. *N,N*–

Dimethylbutylamine (3.222 g, 24.93 mmol) was dissolved in acetonitrile. (3-Chloropropyl)trimethoxysilane (5.450 g, 27.43 mmol) was added slowly, and the resulting solution was stirred at 90 °C for 24 h under a nitrogen atmosphere. After removing solvent under a dynamic vacuum, the crude product was diluted using diethyl ether and washed with water. The organic layer was dried over anhydrous magnesium sulfate, filtered, and concentrated under a dynamic vacuum to obtain a colorless oil with a 72 % yield. ¹H NMR (400 MHz, DMSO-*d*₆, δ): 0.57 (Si-CH₂, 2H), 0.93 (CH₂-CH₃, 3H), 1.29 (CH₂-CH₂-CH₃, 2H), 1.64 (N-CH₂-CH₂, 4H), 2.95-3.05 (N-CH₂, 4H), 3.10-3.30 (N-CH₃, 6H), 3.45-3.53 (Si-O-CH₃, 9H)

3.2.4. Preparation of #AC-coated glass

Microscope glass was precisely cut into dimensions of 2.5 cm × 2.5 cm to serve as the substrate. The glass substrate was immersed in a piranha solution [a mixture of concentrated sulfuric acid (98 wt.%) and hydrogen peroxide solution (30 wt.%) (7:3 v/v)] for 30 min. Following this treatment, the substrate was rinsed with distilled water and ethanol, and dried under a dynamic vacuum to remove residual solvent. #AC was dissolved in ethanol to prepare a 5 wt.% solution, and 0.5 mL of this solution was drop-cast onto the glass substrate. The solvent was allowed to evaporate at room temperature, after which the coated glass was heated at 70 °C for 16 h. The

coated glass was rinsed three times with ethanol and dried under a dynamic vacuum condition at 60 °C. These #AC-coated glass plates were utilized for comprehensive investigations, including the measurement of water contact angle, evaluation of antibacterial activity, and observation of bacterial adhesion behavior.

3.2.5. Antibacterial activity of #AC

Antibacterial activity of #AC was evaluated using the #AC-coated glass through a film attachment test against *Escherichia coli* (*E. coli*; ATCC 700926). To prepare the bacterial suspension, *E. coli* cells were cultured in the nutrient broth solution at 37 °C for 18 h. A representative colony was lifted off with a platinum loop, placed in 30 mL of nutrient broth, and incubated with shaking at 37 °C for 18 h. After washing twice with saline solution (0.9 wt% NaCl solution), they were re-suspended in saline solution to yield 1×10^6 colony forming unit (CFU) mL⁻¹. Bacterial cell concentration was estimated by measuring the absorbance of cell dispersions at 600 nm using U5100 UV-Visible Spectrometer (Hitachi, Japan) and referenced to a standard calibration curve. An optical density of 0.1 at 600 nm is approximately equivalent to 10⁸ CFU mL⁻¹. To evaluate the antibacterial activity of #AC, 0.1 mL of the bacterial suspension was dropped onto the surface of the #AC-coated glass located in a Petri dish and

the glass was covered by sterilized poly(ethylene terephthalate) films (2.4 cm × 2.4 cm). After 24 h at 25 °C, the saline solution (0.9 mL) was poured into the Petri dishes that contained the samples. After vigorous shaking to detach adherent bacteria from the surface, the solution mixture was transferred to a micro tube. The resulting solution was serially diluted and then 0.1 mL of each diluent was spread on the agar plates. Viable bacterial colonies were counted after incubation for 18 h at 37 °C. Each test was repeated at least three times. The antibacterial activity was calculated as follows:

$$\text{Antibacterial activity (\%)} = 100 \times (N_0 - N_i) / N_0 \quad (1)$$

where N_0 is bacterial CFU of blank and N_i is bacterial CFU of tested sample.

3.2.6. Preparation of surface modified ZS with #AC (ZS#AC)

1.000 g of ZS was reacted with #AC at a constant molar amount of 0.02 mmol. For example, 0.0060 g (0.02 mmol) of 4AC was dissolved in 10.0 mL of isopropyl alcohol. Subsequently, ZS (1.000 g) was added into this #AC solution, and the resulting mixture was stirred at 60 °C for 24 h. After removing the solvent under a dynamic vacuum condition, the crude product was heated at 120 °C for 24 h. The resulting glass powder was denoted as ZS#AC to specify #AC used for surface modification.

3.2.7. Antibacterial activity of ZS and ZS#AC

A shake-flask method was employed to comparatively evaluate the antibacterial activity of ZS and ZS#ACs [14]. To prepare a well-suspended solution containing either ZS or ZS#ACs, 1 mg of the respective powder was added to 2 mL of saline solution, followed by sonication for 1 h. The stock solutions were diluted with saline solution to obtain three distinct concentrations of 0.05, 0.07, and 0.1 mg mL⁻¹. For example, to obtain a 0.1 mg mL⁻¹ ZS solution, 1 mL of the ZS stock solution was added to a well of a 6-well plate, followed by the addition of 3 mL of saline solution and 1 mL of bacterial suspension. After inoculation, the suspensions were incubated at 37 °C for 18 h with shaking at 200 rpm. The resulting solution (0.1 mL) was spread on the agar plates, and viable bacterial colonies were counted after incubation for 18 h at 37 °C. Each test was repeated at least three times. The antibacterial activity in terms of log reduction was calculated as follows:

$$\text{Antibacterial activity (log)} = \log (N_0 / N_i) \quad (2)$$

where N_0 is bacterial CFU of blank and N_i is bacterial CFU of tested sample.

3.2.8. Preparation of polypropylene composites containing ZS and ZS#ACs (PZS_x and PZ#A_x)

Polypropylene (PP) composites containing ZS and ZS#AC are denoted as PZS_x and PZ#A_x, respectively, where x represents the weight percentage of

the respective filler relative to PP. Prior to usage, ZS and ZS#ACs were thoroughly dried under a dynamic vacuum at 60 °C to prevent water adsorption. The following is the representative procedure for the preparation of PZS0.5. ZS (30.0 mg, 0.5 wt.% to PP) was placed in a 250 mL round-bottom flask and dispersed in *p*-xylene (60.0 mL) by sonication for 30 min. Subsequently, 6.00 g of PP was added to the ZS suspension, and the mixture was stirred at 140 °C for 2 h. The resulting solution was poured into a glass Petri dish and heated at 150 °C for 1 h to evaporate the solvent, followed by drying at 60 °C under a dynamic vacuum overnight. To obtain PP composites with well-dispersed fillers, the product was ground into a powder form, followed by compression molding at 200 °C for 3 min. The preparation of other PP composites followed the same procedure with varying filler type and content. A PP film without any filler was also prepared using the same procedure except the preparation of a filler suspension.

3.2.9. Antibacterial activities of PZS_x and PZ#A_x

The antibacterial activities of PP composite films were investigated using a film attachment test against *E. coli*. 0.4 mL of the bacterial suspension (10^6 CFU mL⁻¹) was dropped on the surface of the polymer films (5 cm × 5 cm) located in a Petri dish and the films were covered by sterilized

polyethylene (PE) films (4 cm × 4 cm). After 24 h at 35 °C, the samples were placed into sample bags (Whirl-Pak bags) with 10 mL of SCDLP broth and pummelled in a stomacher (Bagmixer 400 CC, Interscience) for 5 min at a speed of 6 stokes sec⁻¹. After vigorous shaking to detach adherent bacteria from the film, the solution mixture was transferred to a micro tube. The resulting solution was serially diluted and then 0.1 mL of each diluent was spread on the agar plates. Viable bacterial colonies were counted after incubation for 18 h at 37 °C. Each test was repeated at least three times. The antibacterial activity of polymer composites was calculated using the equation (1).

3.2.10. Antiviral activities of PP, PZS2, and PZ6A2

MS2 coliphage (ATCC 15597-B1) was used with *E. coli* C3000 (ATCC 15597) as a host. The host *E. coli* was cultivated in tryptone broth containing tryptone (10 g L⁻¹), NaCl (8 g L⁻¹), yeast extract (1 g L⁻¹), glucose (1 g L⁻¹), 2 mM CaCl₂, and thiamine (0.01 g L⁻¹). MS2 was inoculated in the suspension of host *E. coli*. After incubating for 18 h at 37 °C, the mixture of *E. coli* and MS2 was centrifuged at 3000 *g* for 15 min, and the supernatant was filtered with a 0.22 µm PTFE syringe filter to collect MS2 particles. The population of MS2 was determined by the plaque assay method using media of the top and bottom double layer containing

0.5 % and 1.5 % of agar. The stock solution of MS2 ($\sim 10^{11}$ PFU mL⁻¹) was stored at 4 °C until use. To evaluate the antiviral activities of PP and PP composites, 0.1 mL of the viral suspension ($\sim 10^8$ PFU mL⁻¹) was dropped onto the surfaces of the polymer films (2 cm × 2 cm) located in a Petri dish. The films were covered using clear PE films having the same size to ensure full contact. After 24 h at 37 °C, 2.4 mL of PBS was poured into the samples, followed by vigorously shaking to detach viral particles adhered on the films. A 1.0 mL aliquot was withdrawn, filtered through a 0.22 μ m PTFE syringe filter, and serially diluted with PBS to quench the reaction. The samples were spread on agar plates to count the viable viral particles through the plaque assay.

3.2.11. Biocompatibility Assay of PP, PZS2, and PZ6A2

Human keratinocytes cell (HaCaT cells, elabscience, US) was cultured in minimum essential medium (MEM, Gibco, Massachusetts), containing 10 % fetal bovine serum (FBS, Corning, New York) and 1 % Penicillin-streptomycin (Gibco, Massachusetts) at 37 °C in an atmosphere containing 5 % CO₂ with 95 % relative humidity. All the samples were sterilized by UV irradiation before the biocompatibility assay. For Live/Dead assay to investigate the biocompatibility, the sterilized samples were added into each well of a 24-well plate containing cultured cells (2×10^4 cells in each well)

and incubated in direct contact with the cells for 24 h and 72 h. As a negative control, tissue culture grade polystyrene (TCP) was used. After incubation, the cell viability was measured by observing the stained cells using the fluorescence microscope (EVOS Cell Imaging Systems, Thermo Fisher Scientific), where the cells were stained with Live/Dead Viability/Cytotoxicity Kit that contains calcein-AM and ethidium homodimer-1 (EthD-1).

3.3. Results and Discussion

3.3.1. Synthesis and characterization of ZS#AC

To address the microbial susceptibility of PP, zinc silicate glass powder (ZS) was incorporated as an antimicrobial inorganic filler, based on our previous research demonstrating its superior efficiency through comparative analysis with commercial zinc oxide [9]. It should be noted that ZS utilized in this study exhibits a larger diameter ($1.47\ \mu\text{m}$) compared to its counterpart ($0.34\ \mu\text{m}$) used in the previous study due to the omission of a separation procedure (Figure 3.2a). Despite this size difference, their chemical compositions are nearly close, as confirmed by XRF (Figure 3.2b).

While ZS exhibited remarkable efficiency as an antimicrobial filler for PP in our previous research, its intrinsic characteristics still present prominent challenges in achieving optimal performance of PP composites. The major challenge originates from the gradual depletion of zinc ions, resulting in a decrease in antimicrobial activity, thereby limiting its long-term efficiency. In addition, its hydrophilic surface promotes aggregation due to the incompatibility with hydrophobic PP matrix, potentially causing deterioration of overall performance [15]. As illustrated in Figure 3.1, silanization was employed to overcome these challenges, in a way that incorporates antimicrobial moiety and modulates its surface properties. In

pursuit of these objectives, we designed a series of silane coupling agents, (dimethylalkyl[3-(trimethoxysilyl)propyl]ammonium chloride), denoted as #AC, where # represents the carbon number of alkyl chain. In #ACs, a quaternary ammonium group was incorporated as an antimicrobial component, along with alkyl chains of varying lengths to modulate the surface properties. To prepare #AC series, three silane coupling agents with distinct alkyl groups—butyl, hexyl, and octyl—were synthesized (Figure 3.3a), while commercially available agents, including trimethyl[3-(trimethoxysilyl)propyl]ammonium chloride (1AC) and dimethyloctadecyl[3-(trimethoxysilyl)propyl]ammonium chloride (18AC), were also utilized. The chemical structures of the synthesized #ACs were confirmed by ^1H NMR and FT-IR spectra (Figures 3.3b and 3.4).

The antibacterial activity of #ACs was evaluated using #AC-coated glass through a film attachment method against *E. coli*. As shown in Figure 3.3d and e, 8AC exhibits the strongest antibacterial activity of 98.7 %, followed by 6AC, 18AC, 4AC, and 1AC, with respective antibacterial activities of 97.6 %, 96.4 %, 93.3 %, and 88.2 %. Notably, it demonstrates that the antibacterial activity of #AC depends on its alkyl chain length, wherein the antibacterial activity increases with # until it reaches 8, and subsequently decreases. The effect of alkyl chain length on antimicrobial activity has been widely investigated, demonstrating the presence of optimal alkyl chain

length for effectively disrupting the bacterial membrane [16]. In addition, the effect of alkyl chains on the surface properties was investigated by measuring the water contact angle of #AC-coated glass. As shown in Figure 3.3c, there is a linear relationship between the alkyl chain length and surface hydrophobicity, as the water contact angle of #AC-coated glass increases with #. Therefore, it can be concluded that #AC series can serve as an effective model system to enhance long-term antibacterial activity and modulate the surface properties of ZS, thereby optimizing the overall performances of PP composites, especially antimicrobial activity.

Figure 3.5 illustrates the surface modification of ZS employing the #AC series, and resulting modified ZS is referred to as ZS#AC. Note that a constant molar amount of 0.02 mmol of #ACs was incorporated relative to the unit mass of ZS for modification. TGA curves indicate the incorporation of #ACs into ZS#ACs, as ZS#ACs exhibit smaller char yield values at 700 °C compared to ZS (Figure 3.5b). Furthermore, the char yield of ZS#ACs decreases with #, as the molecular weight of #AC increases with #. The chemical structure of ZS and ZS#ACs was investigated using various spectroscopic analyses, including FT-IR, Raman, XPS, and ^{29}Si MAS NMR (Figures 3.5c-e and 3.6). In FT-IR spectra, both ZS and ZS#ACs comprise characteristic peaks originating from the unique structure of ZS. Specifically, two prominent peaks at 915 cm^{-1} and 502 cm^{-1} correspond to asymmetric

stretching and the rocking mode of Si-O-Si(Na), respectively (Figure 3.5c) [17]. Additionally, a peak at 695 cm^{-1} represents the presence of zinc integrated within the silicate network of both ZS and ZS#ACs [18]. However, a small peak at 1444 cm^{-1} was exclusively observed in FT-IR spectra of ZS#ACs, which corresponds to CH_2 bending mode, indicating the incorporation of #ACs [19]. Moreover, Raman spectra further corroborated the incorporation of #AC onto ZS#ACs, by five distinct peaks located at 1131 cm^{-1} , 1227 cm^{-1} , 1316 cm^{-1} , 1399 cm^{-1} , and 1515 cm^{-1} , where the lowest wavenumber peak corresponds to C-C stretching and the others to CH_2 bending (Figure 3.5d) [20].

Although FT-IR and Raman spectra suggest the successful incorporation of #ACs onto ZS#ACs, additional analyses should be conducted to verify their chemical structures. Therefore, the chemical bonding state of ZS#AC surfaces was examined using their deconvoluted XPS spectra of the Si 2p and O 1s regions (Figures 3.7 and 3.8). In the O 1s region, a peak was observed at around 533 cm^{-1} , corresponding to surface silanol moiety present in both ZS and ZS#ACs. However, after the surface modification using #ACs, the atomic percentage of silanol moiety decreased, indicating its participation in silanization to form covalent bonds (Figure 3.5e) [21]. Furthermore, in the Si 2p region, a peak at 101.6 cm^{-1} corresponding to Si-C/Si²⁺ became more pronounced in ZS#ACs compared to ZS, indicating the

formation of Si-C bonds via silanization [22]. The formation of Si-C bonds was further corroborated by ^{29}Si MAS NMR spectra, where a peak at around -60 ppm corresponding to T_2 species was exclusively observed in ZS#ACs (Figure 3.6). Consequently, these comprehensive analyses clearly confirm the successful surface modification of ZS through silanization using #ACs. Despite the chemical structure modification, the amorphous structure and morphology of ZS were retained after the surface modification, as shown in Figure 3.9.

3.3.2. Antibacterial activities of ZS#AC

To investigate the effect of surface modification on the antibacterial activity, the antibacterial activity of ZS and ZS#ACs was evaluated by a shake-flask method against *E. coli* (Figure 3.10a). At a suspension concentration of 0.10 mg mL^{-1} , no viable bacteria were observed on the agar plates for ZS, ZS1AC, ZS4AC, and ZS6AC, indicating their minimum bactericidal concentration values as 0.10 mg mL^{-1} . In contrast, there were a few bacterial colonies on the agar plates for ZS8AC and ZS18AC, indicating their weaker antimicrobial activity compared to the others. Furthermore, the trend in antimicrobial activity was clearly identified by measuring their antibacterial activity at a suspension concentration of 0.07 mg mL^{-1} , as summarized in Figure 3.10b. Despite the incorporation of

antibacterial quaternary ammonium groups in ZS#ACs, ZS exhibits the strongest antibacterial activity with over a 7-log reduction. The antibacterial activity of ZS#ACs tends to decrease with increasing #, while it was not feasible to calculate the antibacterial activity of ZS#8AC and ZS#18AC due to the considerable number of bacterial colonies observed on their agar plates.

Since the antimicrobial activity of ZS is primarily attributed to the release of zinc ions, the amount of released zinc ions from ZS and ZS#ACs was measured using ICP-MS. In Figure 3.10c, ZS exhibits the largest amount of released zinc ions, whereas the release of zinc ions gradually decreases in ZS#ACs with increasing #. We attribute this slower zinc ion release from ZS#ACs compared to ZS to the presence of the grafted #AC layer. This layer can affect the interaction with water molecules by altering the surface properties of ZS#ACs or physically blocking the direct contact with the aqueous environment [23]. Therefore, the surface properties of ZS and ZS#ACs were investigated by measuring the contact angle values of water and diiodomethane (Figure 3.10e). ZS display its hydrophilic nature, with a small water contact angle value of 23.2° and a large surface energy value of 69.9 mN m^{-1} , which is nearly close to that of water (72 mN m^{-1}) [24]. In sharp contrast, ZS#ACs exhibit significantly larger water contact angle value in the range between 64.0 - 113.6° . Furthermore, the gradual increase in

water contact angle and decrease in surface energy clearly indicates the increase in surface hydrophobicity of ZS#ACs with increasing #, confirming the precise modulation of surface properties through the incorporation of #ACs.

To elucidate the effect of surface properties on the interaction between fillers and water molecules, the dispersion behavior of ZS and ZS#ACs was analyzed by observing the time-dependent absorbance at 300 nm. As shown in Figure 3.10d, the ZS suspension exhibits the largest absorbance value for all times, indicating the strong interaction with water molecules due to its hydrophilic surface. In contrast, ZS#AC suspensions exhibit smaller absorbance values with more rapid decrease over time compared to the ZS suspension. Furthermore, this trend becomes more pronounced as # increases, indicating the poor dispersion of ZS#ACs in aqueous solution due to the increase in surface hydrophobicity. Consequently, it can be concluded that the weaker antimicrobial activity of ZS#ACs is attributed to the increase in surface hydrophobicity resulting from surface modification using #ACs, which inhibits the effective release of zinc ions and physical interaction with bacteria. Still, we believe that ZS#ACs have potential to enhance the antibacterial activity of PP composites, as the PP matrix is inherently hydrophobic, in sharp contrast to the aqueous medium.

3.3.3. Dispersion behavior of ZS#AC in PP matrix

Figure 3.11 summarizes the dispersion behavior of fillers within the PP matrix. This behavior is known to be determined by interactions between fillers and the polymer matrix, and more uniform dispersion of fillers can be achieved through strong interfacial interaction between filler and the polymer matrix [25]. Therefore, the surface modification using #ACs can promote more uniform dispersion of fillers by modulating their surface properties to enhance the interfacial interaction with the PP matrix.

To investigate the effect of surface modification, Hansen solubility parameters (HSP) were employed, as they facilitate comparative evaluation of the interfacial interaction between fillers and the PP matrix. HSP consists of three independent terms: δ_D , δ_P , and δ_H , representing disperse, polar, and hydrogen bonding forces, respectively [26]. HSP values of ZS#ACs are listed in Table 3.2 with a detailed explanation of their calculation using a group contribution method. Figure 3.11a clearly illustrates the interfacial affinity between fillers and PP by plotting their HSP values in a Hansen space with a coordination of $(2\delta_D, \delta_P, \delta_H)$. A grey sphere represents the Hansen sphere of PP with a geometric center of (36, 0, 1) and an interaction radius (R_o) of 6.0, depicting a region of HSP values for materials exhibiting good compatibility with PP such as *p*-xylene [27]. Notably, all ZS#ACs were located within the PP Hansen sphere, indicating their strong interfacial

interaction with PP. However, inorganic fillers with compositions close to ZS show a marked discrepancy in their HSP values compared to PP, suggesting their poor interfacial affinity due to highly hydrophilic surface of ZS [28]. Moreover, the interfacial affinity of fillers with the PP matrix can be quantitatively evaluated by measuring their relative energy difference (RED) value, which is a ratio of R_a and R_o , where R_a is a solubility parameter distance between the filler and the PP matrix in the Hansen space [29]. As shown in Figure 3.11b, ZS analogues exhibit the largest RED value of 4.33, while all ZS#ACs show significantly smaller RED values below 1.00. Moreover, RED values of ZS#ACs gradually decrease with increasing #, indicating stronger interfacial interaction with the PP matrix. Therefore, it can be concluded that the surface modification using #ACs effectively improves the interfacial interaction with the PP matrix by increasing the surface hydrophobicity of fillers. In addition, the dispersion behavior of ZS and ZS#AC suspensions in *p*-xylene provides additional indication of this phenomenon, as ZS#ACs with larger # exhibit more stable dispersion with larger absorbance value over time (Figure 3.11c).

To observe the dispersion behavior of fillers within PP, PP composites were prepared using either ZS or ZS#ACs as fillers, with a fixed filler content of 1 wt.% relative to PP. The resulting PP composites are denoted as PZS1 and PZ#A1, where the suffix 1 indicates the filler content, and ZS and

Z#A represent ZS and ZS#AC, respectively. In Figure 3.12a, all PP composites show high transparency without noticeable aggregation of filler. Therefore, cross-sectional SEM images of the fractured PP composites were precisely observed to investigate the interfacial interaction between fillers and the PP matrix (Figure 3.11d). In PZS1, debonding between ZS and the PP matrix was observed with a large gap between them, indicating their poor interfacial interaction [30]. Although debonding between a filler and the PP matrix was also observed in PZ1A1 and PZ4A1, the gap between them became much smaller than that observed in PZS1. Notably, in PZ6A1, PZ8A1, and PZ18A1, the filler was fully embedded in the PP matrix, demonstrating the strong interfacial interaction between these fillers and the PP matrix. Consequently, these SEM images represent the stronger interfacial interaction of ZS#AC with the PP matrix compared to ZS, especially for ZS6AC, ZS8AC, and ZS18AC.

Finally, the dispersion behavior of fillers within polymer composites was analyzed using XRM. Figure 3.11f exhibits reconstructed XRM images of polymer composites, where the PP matrix was omitted to clearly distinguish filler particles. In PZS1, a large-sized particle was observed, potentially due to the aggregation of ZS, indicating its poor dispersity. Furthermore, XRM images reveal a more uniform dispersion of fillers in PZ6A1, PZ8A1, and PZ18A1 compared to the others, attributed to stronger interfacial interaction

between these respective fillers and the PP matrix. The particle volume distribution determined by XRM supports this dispersion behavior, where the percentage of particles with smaller volume is largest for PZ18A1, followed by PZ8A1, PZ6A1, PZ4A1, PZ1A1, and PZS1 (Figure 3.11e). Consequently, it can be concluded that the surface modification using #AC with larger # can achieve more uniform dispersion of the filler within the composite by promoting the interfacial interaction between filler and the PP matrix.

The effect of filler dispersion behavior on the overall performance of polymer composites was investigated through comprehensive analyses on their thermal and mechanical properties using TGA, DSC, and tensile testing (Figures 3.12 and 3.13). In Figure 3.12b, the maximum degradation temperature (T_{dmax}), at which the 1st derivatives of TGA curves exhibit the maximum value, is higher for PZ#A1 than PZS1. Furthermore, T_{dmax} tends to increase with increasing #, indicating improved thermal stability resulting from a more uniform dispersion of fillers. In DSC cooling curves, an exothermic peak appears in the range of 115-125 °C, originating from the crystallization of PP (Figure 3.12c). As shown in Figure 3.12d, the crystallization onset temperature (T_{oc}) of PZ#A1 is higher than that of PZS1, indicating stronger nucleating effect of ZS#ACs. Furthermore, in PZ#A1s, T_{oc} tends to increase with increasing #, as the more uniform dispersion

increases the active surface area of fillers, promoting nucleation [31]. Since the nucleation effect of fillers can promote the crystallization of PP, a larger size or number of PP crystallites can be formed, resulting in an increase in ΔH_c . Indeed, ΔH_c values of PP composites tend to increase in PZ#A1s with increasing # (Figure 3.12d). In addition, POM images illustrate the crystallization behavior of polymer composites, with a clear distinction of PP crystallites identified as white spherulites. As shown in Figure 3.12e, PP crystallites were more densely distributed in PZ6A1, PZ8A1, and PZ18A1 compared to the others. This is attributed to the more uniform dispersion of these respective fillers, thereby promoting their nucleating effect.

The mechanical properties of PP composites were evaluated to investigate the effect of filler dispersity. In stress-strain curves of PZS1 and PZ#A1s, PZ#A1s exhibit larger tensile strength value than PZS1 (Figure 3.13a). Furthermore, in PZ#A1s, both tensile strength and Young's modulus values increase with increasing # (Figure 3.13b-c). This underscores a crucial role of uniform filler dispersion in improving the mechanical strength of polymer composites. Besides, strong interfacial interaction between fillers and the PP matrix can effectively impede debonding between them, thereby contributing to the mechanical robustness of the composites [32]. In addition, PZ#A1s generally exhibit greater elongation at break value compared to PZS1, potentially attributed to more uniform dispersion of

fillers. However, unlike mechanical strength values, elongation at break exhibits a subtle decrease as # exceeds 6 in PZ#A1s. This can be associated with the robust interfacial adhesion of densely distributed fillers, which in turn hampers the plastic deformation of PP. Consequently, it can be concluded that the dispersion behavior of fillers, determined by surface modification, exerts significant impact on overall performance of PP composites.

3.3.4. Antimicrobial activities of PZ#A_x composites

As shown in Figure 3.14a, the antibacterial activities of PZS1 and PZ#A1s were evaluated against *E. coli* using a film attachment test to investigate the effect of surface modification using #ACs. Intriguingly, the antibacterial activity trend in PP composites presents a stark contrast to that observed in antibacterial activity of the filler itself. Although ZS exhibits the strongest antibacterial activity among these fillers, PZS1 exhibits the weakest antibacterial activity among polymer composites, with a value of 88.43 % (Figure 3.14b). Conversely, all PZ#A1s demonstrate stronger antibacterial activity ranging from 93 % to 100 %, surpassing PZS1. In PZ#A1s, antibacterial activity increases with increasing # until it reaches 6, where PZ6A1 and PZ8A1 exhibit the maximum antibacterial activity of 100 %. Subsequently, a slight decrease is observed in PZ18A1, with very

few bacterial colonies observed on the agar plate.

To understand the antibacterial behavior in PP composites, the release of zinc ions from PP composites was measured by ICP-MS. In Figure 3.14c, all PZ#A1s exhibits a larger amount of released zinc ions compared to PZS1, contrary to the results observed in their fillers. Furthermore, the amount of released zinc ions tends to increase with increasing # in PZ#A1s. This demonstrate that more uniform dispersion of fillers can promote the release of zinc ions by increasing their surface area, ultimately improving antibacterial activity of polymer composites. Furthermore, the uniform dispersion of fillers promotes the interaction between fillers and bacteria, thereby efficiently disrupting bacterial outer membrane through physical damage and electrostatic attraction with quaternary ammonium moiety on ZS#AC surfaces. In addition, a subtle decrease of zinc ion release was observed in PZ18A1, potentially attributed to long alkyl chain covering its surface, which might impede the interaction with water molecules.

Besides antibacterial activity, effective prevention of bacterial adhesion is also crucial, as adhered bacteria can promote further attachment and proliferation of bacteria, ultimately leading to the formation of biofilm, which can significantly diminish antibacterial activity of PP composites [33]. Therefore, bacterial adhesion behavior of PP and PP composites was observed using CLSM, allowing for a clear distinction between live and

dead bacteria through green and red staining, respectively (Figure 3.14d). A considerable number of live bacteria was observed on the PP surface, indicating its susceptibility to bacterial contamination. In contrast, PP composites exhibit a significant smaller number of bacteria on their surfaces, indicating effective prevention of bacterial adhesion. Furthermore, the stronger antibacterial activity of PZ#A1s was identified by a much smaller ratio of live bacteria compared to PZS1, as listed in Table 3.3. Specifically, only dead bacteria were observed on PZ6A1 and PZ8A1, indicating their superior antibacterial activity of these composites. However, the total number of adhered bacteria is the smallest for PZS1, and it tends to increase with increasing # in PZ#A1s.

To elucidate the bacterial adhesion behavior, the surface properties of PP and PP composites were investigated by measuring their water contact angle and ζ -potential values. In Figure 3.14e, PP demonstrates its inherent hydrophobic nature by displaying a large water contact angle value of 103.15°. A sharp decrease in the water contact angle value was observed in PZS1, with a value of 91.67°, due to the incorporation of hydrophilic ZS. The water contact angle values for PZ#A1 are larger than that of PZS1, as the surface modification using #ACs increases the surface hydrophobicity of fillers. Furthermore, as the surface hydrophobicity of ZS#ACs increases with increasing #, a gradual increase in water contact angle values was

observed in PZ#A1s. Consequently, bacterial adhesion can be promoted on the PZ#A1 surface with larger # due to stronger hydrophobic interaction with bacteria [34]. In sharp contrast, the highly hydrophilic ZS facilitates the formation of a hydration layer, effectively preventing bacterial adhesion on the PZS1 surface, as confirmed in CLSM images.

Since *E. coli* displays an overall negative potential due to its abundant hydroxyl and phosphate groups in its outer membrane, the negative charge of a surface can effectively prevent bacterial adhesion through electrostatic repulsion [9]. In Figure 3.14f, PP exhibits a surface ζ -potential of -6.08 mV, while PZS1 displays a much more negative ζ -potential of -27.27 mV due to the negative surface charge of ZS. The surface ζ -potentials of PZ#A1s are less negative than that of PZS1. Specifically, the surface ζ -potentials are -19.96 mV, -19.88 mV, -22.33 mV, -21.57 mV, and -15.89 mV for PZ1A1, PZ4A1, PZ6A1, PZ8A1, and PZ18A1, respectively. This can be attributed to the incorporation of #AC with a positively charged quaternary ammonium group, compensating for the negative charge of ZS. Nonetheless, since PZ#A1 exhibits lower ζ -potential compared to PP, the introduction of ZS#AC may result in reduced bacterial adhesion relative to the PP. Based on these findings, the surface modification using #AC can be regarded as a promising strategy to improve both antibacterial activity and filler dispersion of PP composites. However, it should be noted that there is an

optimal length of alkyl chain to achieve robust antibacterial activity, while simultaneously preventing bacterial adhesion. In this regard, ZS6AC emerged as the most promising candidate to optimize the overall performance of PP composites.

To further evaluate the efficiency of ZS6AC as an antimicrobial filler, PP composites with ZS and ZS6AC were prepared by varying their content, and the resulting PP composites are denoted as PZS x and PZ6A x , where x represents the filler content. As shown in Figure 3.15a-b, the antibacterial activity of PZS x and PZ6A x against *E. coli* were evaluated using a film attachment method. In both PZS x and PZ6A x , the antibacterial activity increases as the filler content increases. Remarkably, PZ6A x s exhibit superior antibacterial activity compared to PZS x s across the entire range of x . Specifically, PZ6A1 exhibits a maximum antibacterial activity of 100 %, while PZS1 shows significantly weaker antibacterial activity of 88 %. Furthermore, in PZS x , the maximum antibacterial activity was achieved by incorporating ZS more than 4 wt.%, highlighting the superior efficiency of ZS6AC. We speculate that this superior efficiency of ZS6AC as an antibacterial filler can be attributed to its more uniform dispersion within the PP matrix, thereby promoting the release of zinc ions and interaction with bacteria, simultaneously.

Since the interaction with bacteria mainly occurs at the surface of

polymer composites, their surface was observed using the backscattered electron (BSE) images, enabling a clear distinction of fillers from the PP matrix (Figure 3.16) [35]. BSE images of PZ6Axs demonstrate the more uniform and dense distribution of ZS6AC particles, distinguished by their distinct bright color. In sharp contrast, BSE images of PZSxs exhibit the poor dispersion of ZS particles, which are sparsely distributed with irregular sizes. Moreover, particles with a diameter above 1.5 μm (average diameter of ZS) were exclusively observed in PZS10, indicating the presence of aggregates in PZS10 due to the poor interfacial interaction between ZS particles and the PP matrix (Figure 3.17). Consequently, as shown in Figure 3.15c, PZ6Axs exhibit larger area fraction of fillers compared to PZSxs across the entire range of x . This can contribute their superior antibacterial activity, in a way that promotes the interaction with bacteria and release of zinc ions, simultaneously. Indeed, as shown in Figure 3.15d, PZ6Axs released higher concentration of zinc ions compared to PZSxs across the entire range of x .

Zinc ions have been acknowledged for their ability to deactivate not only bacteria but also viruses, as they can influence biological processes, including the inhibition of enzyme activity or viral replication [36]. Therefore, the antiviral activity of PZS2 and PZ6A2 was evaluated against bacteriophage MS2, a representative non-enveloped virus. As shown in

Figure 3.15e, both PZS2 and PZ6A2 exhibit stronger antiviral activity compared to PP; indicating the antiviral activity of both ZS and ZS6AC. However, PZ6A2 exhibits superior antiviral activity with a 3.50-log reduction compared to PZS2 (1.91-log reduction), which is primarily attributed to the more uniform dispersion of ZS6AC in the PP matrix, leading to the effective release of zinc ions. Consequently, it further highlights the superior efficiency of ZS6AC as an antimicrobial filler for PP composites.

Along with strong antimicrobial activity, biocompatibility is essential for the versatile utilization of PP composites, including packaging and biomedical applications [37, 38]. Therefore, biocompatibility of PZS2 and PZ6A2 was evaluated through the Live/Dead assay using HaCaT (human keratinocytes) cells, with TCP (tissue culture grade polystyrene) as the negative control. Figure 3.15f demonstrates good biocompatibility of both PZS2 and PZ6A2, displaying no significant difference from neat PP and the control. Although a few dead cells were observed on the surfaces of PZS2 and PZ6A2, the percentage of dead cell remained below 1.5 % for both after 1 day of incubation (Table 3.4). Moreover, the percentage of dead cell of these PP composites even decreased after 3 days of incubation, highlighting their good biocompatibility. Interestingly, even after the incorporation of cationic quaternary ammonium groups onto the ZS6AC surface,

biocompatibility was well retained, despite concerns raised in some reports regarding the cytotoxicity of cationic compounds [39]. This can be attributed to the immobilization of the quaternary ammonium groups via covalent bonds, preventing the release of the quaternary ammonium compounds, thereby minimizing the potential toxicity [40-42]. Therefore, silanization using #ACs can be regarded as an efficient strategy to prepare a superior antimicrobial filler with excellent biocompatibility.

3.3.5. Long-term antibacterial activity of PZ6A_x composites

Figure 3.18a-b demonstrate the long-term antibacterial activity of PZS2 and PZ6A2. These samples underwent immersed in deionized water for various durations to promote the consumption of zinc ions. Since the continuous depletion of zinc ions leads to a decrease in the antibacterial activity of inorganic fillers, PZS2 exhibits a gradual decrease in its antibacterial activity with longer immersion periods. Specifically, its antibacterial activity decreases from 99.9 % to 46.2 % after three days of immersion. However, despite the limited release of zinc ions, the antibacterial activity of PZ6A2 was retained , as confirmed by its strong antibacterial activity above 90% after 5 days of immersion. Furthermore, even after 28 days, PZ6A2 still exhibits antibacterial activity above 50%, whereas PZS2 completely loses its antibacterial effectiveness.

The time-dependent release of zinc ions from these composites was measured using ICP-MS, as shown in Figure 3.18c. Both PZS2 and PZ6A2 show a gradual decrease in zinc ion release, with their cumulative zinc ion release increasing slowly after certain durations. Notably, PZS2 hardly released zinc ions after 2 days, resulting in a sharp decrease in its antibacterial activity to 72.5 % after 2 days of immersion. In contrast, PZ6A2 released a higher concentration of zinc ions and for a longer duration compared to PZS2, attributed to the more uniform dispersion of ZS6AC. After 3 days of immersion, PZ6A2 also demonstrates the depletion of zinc ions with a subtle increase in its cumulative release of zinc ions. However, despite the consumption of zinc ions, PZ6A2 still exhibits strong antibacterial activity after 3 days of immersion, indicating the effect of quaternary ammonium group on its persistent antibacterial activity.

The proposed antibacterial mechanisms for PZ6A2 are illustrated in Figure 3.18d. Since the antibacterial activity of ZS primarily relies on the release of metal ions, its antibacterial activity significantly decreases over time due to the consumption of zinc ions [43, 44]. However, ZS6AC maintains consistently robust antibacterial activity due to its synergistic collaboration of bactericidal quaternary ammonium group and the release of zinc ions. Specifically, in initial stage, PZ6AC efficiently eradicate the bacteria through the release of zinc ions, promoted by the uniform

dispersion of fillers. Afterwards, despite the small amount of released zinc ions, the quaternary ammonium group contributes to its strong antibacterial activity through electrostatic attraction with negatively charged bacteria, resulting in effective disruption of bacterial cell membrane. Consequently, ZS6AC emerges as an effective antibacterial filler for PP composites, offering robust and persistent antibacterial activity while maintaining excellent biocompatibility.

3.3.6. Mechanical and thermal properties of PZ6A_x composites

The mechanical properties of PZS_x and PZ6A_x were evaluated through tensile testing to investigate the effect of ZS and ZS6AC as fillers (Figure 3.19). As depicted in Figure 3.19a-b, the tensile strength and Young's modulus of both PZS_x and PZ6A_x exhibited significant improvements compared to the PP without any fillers. Conversely, the elongation at break values of these composites showed a reduction. This is consistent with the previous studies, which suggests that the incorporation of rigid nanofillers typically enhances the mechanical properties of polymer nanocomposites by increasing stiffness and strength, while potentially reducing ductility [9, 45]. The enhancement of mechanical properties of polymer composite is highly dependent on the dispersion stability of filler and the interfacial interactions between the filler and the polymer matrix [46]. Therefore, for PP composites,

studies have focused on modifying the surfaces of poorly compatible fillers such as ZS, aiming to enhance their interaction with the PP matrix [47, 48]. As demonstrated in Figures 3.11 and 3.13, introducing alkyl chains to the ZS surface significantly enhances its compatibility with PP, leading to greater improvements tensile strength and Young's modulus at the same filler content (1 wt.%). In Figure 3.19c-d, when the filler content was increased from 0.5 to 10 wt.%, ZS6AC, which was more uniformly dispersed within the PP matrix, consistently enhanced the tensile strength and Young's modulus of the PP composites to a greater extent than ZS. The tensile strength of PZS_x increased with the filler content up to 2 wt.%, but decreased sharply when the ZS content exceeded 4 wt.%. In contrast, PZ6A_x exhibited only a minor decline in tensile strength, even when the ZS6AC content was increased to 10 wt.%. This is because PZ6A_x exhibited only minor agglomeration in the PZ6A10 and maintained uniform filler dispersion, in contrast to PZS_x, where significant filler aggregation is observed starting from PZS4 (Figure 3.20). Young's modulus of both PZS_x and PZ6A_x continued to increase with filler content but exhibited a smaller improvement when the filler content exceeded 4 wt.%. To enhance the Young's modulus of the PP composite, an increase in the interfacial area between the filler and the polymer matrix is necessary. Since filler agglomeration restricts this, PZ6A_x demonstrated a higher young's modulus

value [49]. Therefore, ZS6AC, which is barely aggregated and has strong interactions with PP, contributed to a greater increase in Young's modulus.

In Figure 3.19e, as the rigid filler content increased in both PZSx and PZ6Ax, the mobility of the polymer chains was restricted, leading to a decrease in the elongation at break. When the filler content was identical, PZ6Ax exhibited greater elongation at break values compared to PZSx. As demonstrated in Figure 3.11d, this enhancement is attributed to the hydrophobic alkyl chains, which improve the interfacial interaction between the filler and the PP matrix [50]. During deformation at a constant strain rate, fractures in polymer composites primarily occur through debonding at the interface between polymer matrix and rigid fillers [51]. Agglomerations of fillers or associated defects serve as stress concentrators, thereby leading to the formation of cracks [52, 53].

As demonstrated in Figure 3.12, it has been established that ZS and ZS#AC function as nucleating agents in the PP matrix, thereby significantly influencing the crystallization behavior of PP composites. Both ZS and ZS#AC enhance the crystallinity of PP and facilitate the formation of β -phase crystallites. The increased crystallinity of the PP composites can contribute to improved mechanical properties of semi-crystalline PP [54]. DSC cooling curves of PZSx and PZ6Ax indicate that the crystallization temperature increased with increasing contents of the nucleating agents

(Figure 3.21a-c). Notably, ZS6AC, which exhibited uniform dispersion within the PP matrix, induced a greater increase in T_{oc} compared to ZS, suggesting the formation of more crystallization nuclei [55]. The enthalpy of crystallization (ΔH_c) of the PZS x and PZ6A x exhibited an initial increase and a subsequent decrease with increasing filler content. At low filler content (up to 1 wt.%), the ΔH_c increases due to the nucleating effect of the introduced fillers, whereas at high filler content (from 2 wt.%), it decreases as the rigid filler lowers the mobility of the PP chain and hinders the crystal formation [56]. In PZS x , where more filler agglomeration was observed, the ΔH_c sharply decreased beginning at a filler content of 4 wt.%. Crystallinity, as determined through XRD analysis, similarly presented an initial increase followed by a decrease, aligning with the changes observed in the ΔH_c (Figure 3.22b). From the second heating curves of DSC, the T_m was observed in the sequence of PZ6A x , PZS x , and PP (Figure 3.21d-f). This suggests that ZS6AC provided a more effective nucleation effect compared to ZS, promoting the formation of larger PP crystallites during the cooling stage. The presence of a small shoulder peak around 150 °C in both PZS x and PZ6A x indicates the formation of β -phase crystallites. In Figure 3.22a, this was further corroborated by the detection of the peak corresponding to the 300 plane ($2\theta = 15.9^\circ$) of β -phase crystallites in the XRD patterns [57]. The variation in ΔH_m with increasing filler content in PZS x and PZ6A x

correspond with the changes observed in ΔH_c , further demonstrating the impact of ZS and ZS6AC on the crystallization behavior of the composites.

In Figure 3.23, the influence of ZS and ZS6AC on the thermal stability of the PP composites was evaluated. As thermal decomposition progressed with a gradual temperature increase up to 700 °C, complete decomposition of PP occurred. In contrast, ZS and ZS6AC exhibited a minimal loss of mass, not exceeding 2 % at this temperature (Figure 3.5b), demonstrating their high thermal stability. Consequently, the char yield of PZS_x and PZ6A_x at 700 °C represent the amount of fillers in composites. With an increase in filler content, the char yield also increased, confirming the successful composite fabrication (Figure 3.23a-b). The degradation onset temperature ($T_{d5\%}$) and the maximum degradation temperature (T_{dmax}) showed an increase with the incorporation of the fillers (Figure 3.23c). This suggests that the introduction of ZS and ZS6AC significantly enhances the thermal stability of the composites. Notably, ZS6AC, which is more evenly distributed within the PP matrix, effectively retards the thermal decomposition of the polymer even with the introduction of small quantities [58].

3.4. Conclusions

In this study, we developed an antimicrobial filler for PP composites, achieving superior antimicrobial activity while maintaining excellent biocompatibility through surface modification of ZS. The surface modification using #ACs delicately modulates the surface properties of fillers, thereby significantly affecting their dispersion behavior within the PP matrix. ZS#ACs demonstrate more uniform dispersion within the PP matrix, leading to the enhanced antibacterial activity of PP composites by promoting the release of zinc ions. Notably, ZS6AC achieves a PP composite with the most robust antibacterial activity, while effectively preventing bacterial adhesion, suggesting the presence of an optimal alkyl chain length. The superior efficiency of ZS6AC as an antimicrobial filler was confirmed through comparative evaluation with ZS, demonstrating stronger antimicrobial activity against both *E. coli* and MS2, while maintaining excellent biocompatibility. Furthermore, PZ6A2 consistently maintained antibacterial activity, retaining 98.9 % activity even after immersed in deionized water for 3 days, due to the presence of the quaternary ammonium groups on its surface. Additionally, ZS6AC significantly enhances the overall performance of PP composites, including thermal stability and mechanical strength, attributed to its more uniform

dispersion. We believe that this study offers an insight into the effective strategy to achieve both uniform dispersion and sustainable antimicrobial activity in PP composites with inorganic fillers. Further research using various commercial polymers, such as poly(lactic acid), poly(ethylene terephthalate), and poly(styrene) should be conducted to explore the versatility of this strategy and evaluate the effectiveness of ZS6AC in different matrices.

3.5. References

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Table 3.1. Chemical composition of precursor materials for preparing ZS.

Precursor	ZnO	SiO ₂	K ₂ CO ₃	CaCO ₃	Na ₂ CO ₃	B ₂ O ₃
wt.%	35	34	10	9	10	2

Table 3.2. Hansen solubility parameters of ZS#ACs.

	δ_D (MPa ^{0.5})	δ_P (MPa ^{0.5})	δ_H (MPa ^{0.5})c
ZS1AC	15.90	3.89	0.43
ZS4AC	16.08	3.07	0.38
ZS6AC	16.17	2.69	0.36
ZS8AC	16.23	2.40	0.34
ZS18AC	16.42	1.55	0.27

Table 3.3. The number of live and dead bacteria on PP, PZS1, and PZ#A1s.

Number of bacteria	Live bacteria ^a	Dead bacteria ^a	Total
PP	495	8	503
PZS1	8	7	15
PZ1A1	5	16	21
PZ4A1	5	18	23
PZ6A1	0	36	36
PZ8A1	0	93	93
PZ18A1	8	79	86

^aThe number of bacteria is measured using image J program.

Table 3.4. The number of live and dead HaCaT cells on Control, PP, PZS2, and PZ6A2 surfaces.

	Day 1		Day 3	
	Live cells	Dead cells (percentage %) ^a	Live cells	Dead cells (percentage %) ^a
Control	352	0 (0)	1434	7 (0.5)
PP	318	2 (0.6)	757	1 (0.1)
PZS2	339	5 (1.5)	602	1 (0.2)
PZ6A2	442	6 (1.4)	749	5 (0.7)

^aThe percentage of dead cells was calculated for a comparative evaluation of biocompatibility.

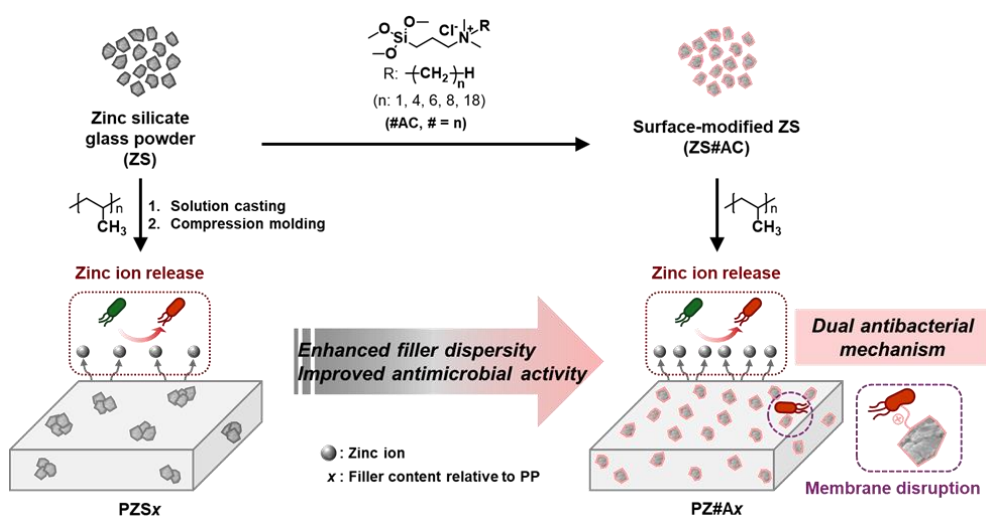


Figure 3.1. Schematic illustrations for the preparation of ZS#AC using silanization and the fabrication of PZ#Ax composites.

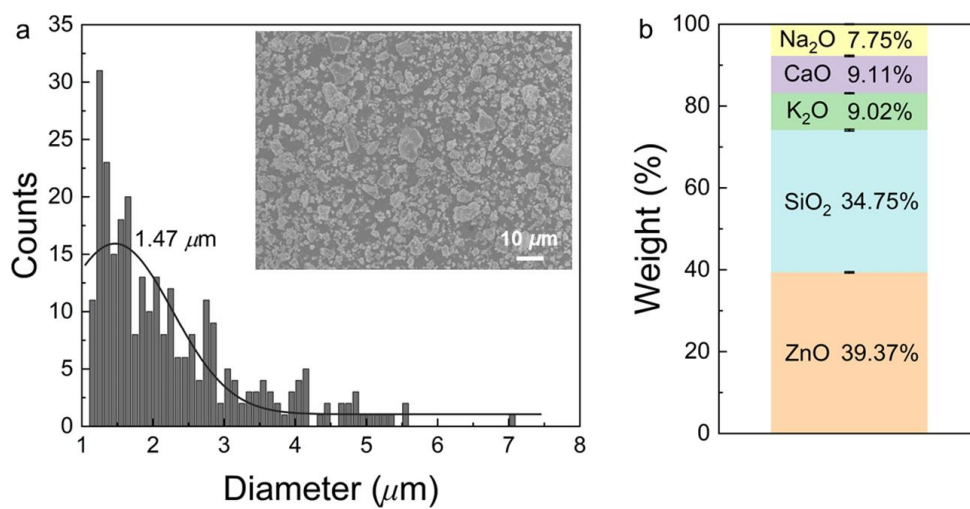


Figure 3.2. (a) Size distribution of ZS determined by SEM image (inset).
(b) Chemical composition of ZS determined by XRF.

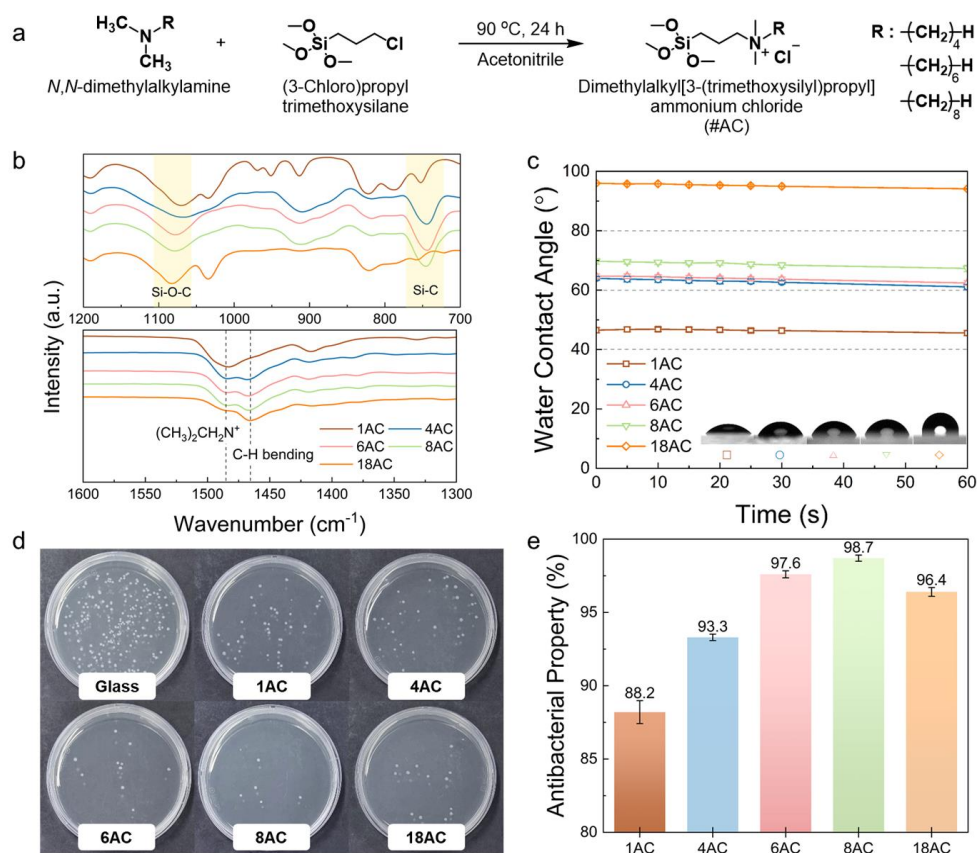


Figure 3.3. (a) Synthetic route to #ACs. (b) FT-IR spectra of #ACs in the range of 1200–700 cm^{-1} (top) and 1600–1300 cm^{-1} (bottom). (c) Water contact angles of #ACs. (d) Photographic results demonstrating antibacterial activity of #ACs against *E. coli*. (e) Calculated antibacterial activity of #ACs.

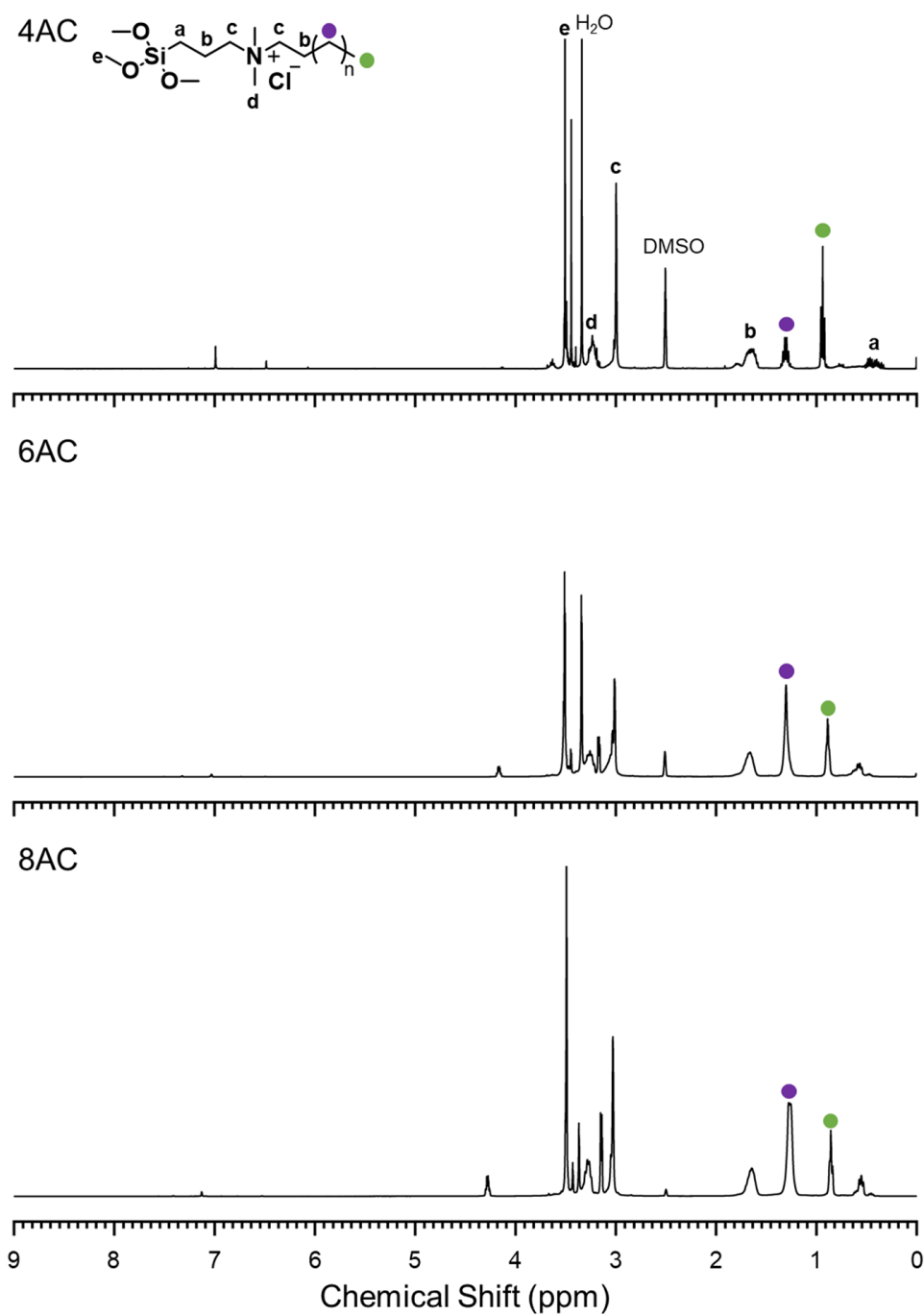


Figure 3.4. ^1H NMR spectra of 4AC, 6AC, and 8AC.

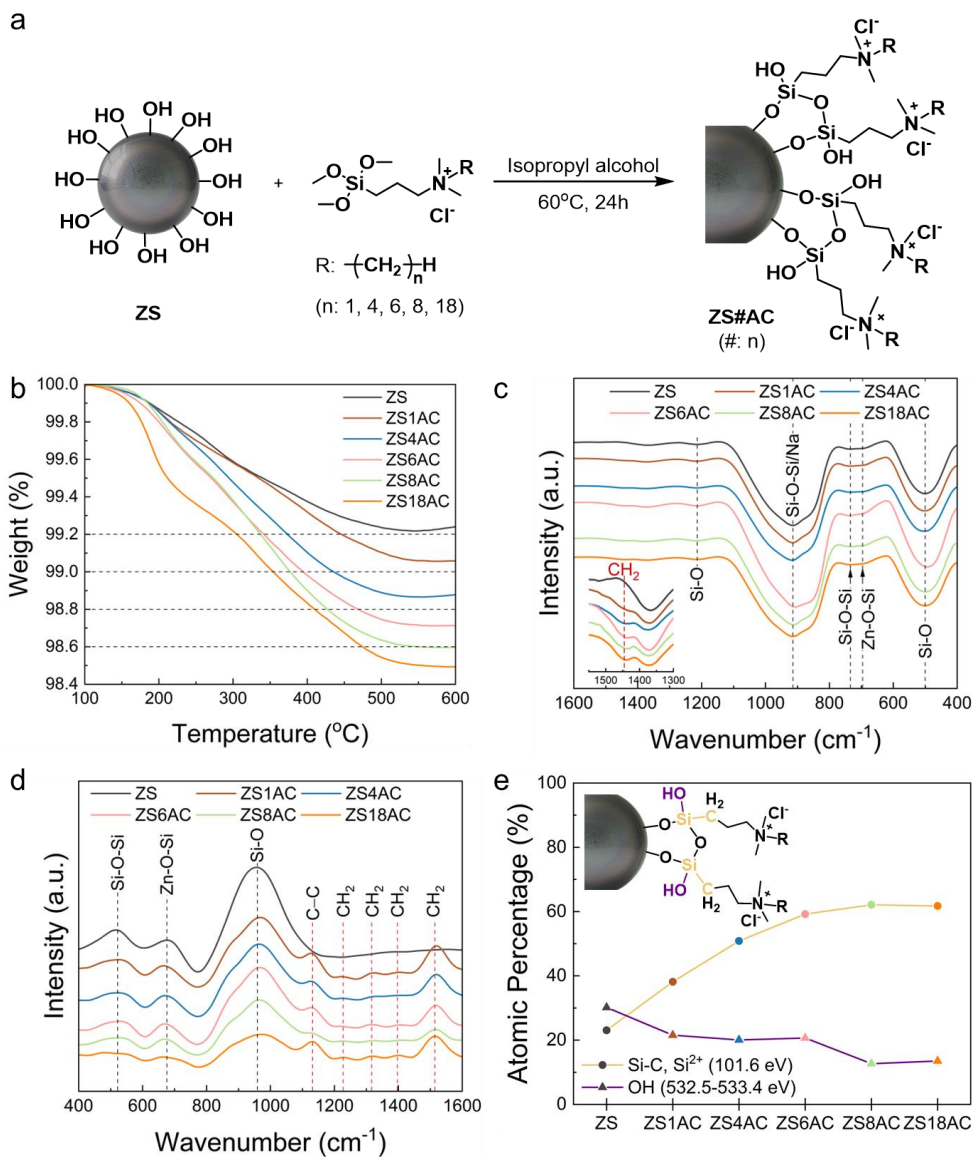


Figure 3.5. (a) Schematics of ZS surface modification using #AC. (b) TGA curves of ZS and ZS#ACs. (c) FT-IR and (d) Raman spectra of ZS and ZS#ACs. (e) Atomic percentage of Si-C and Si-OH species in ZS and ZS#ACs determined by XPS.

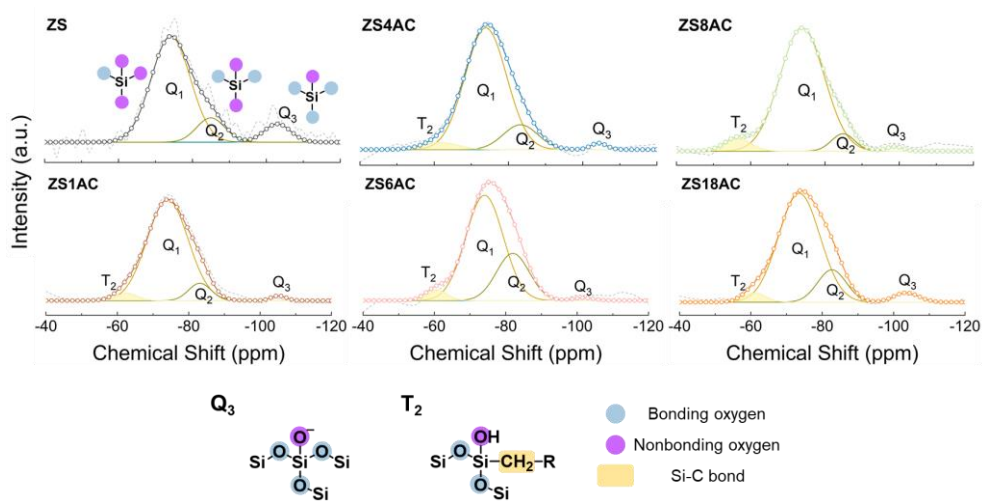


Figure 3.6. ^{29}Si MAS NMR spectra of ZS and ZS#ACs. The molecular structures of Si coordination are depicted below.

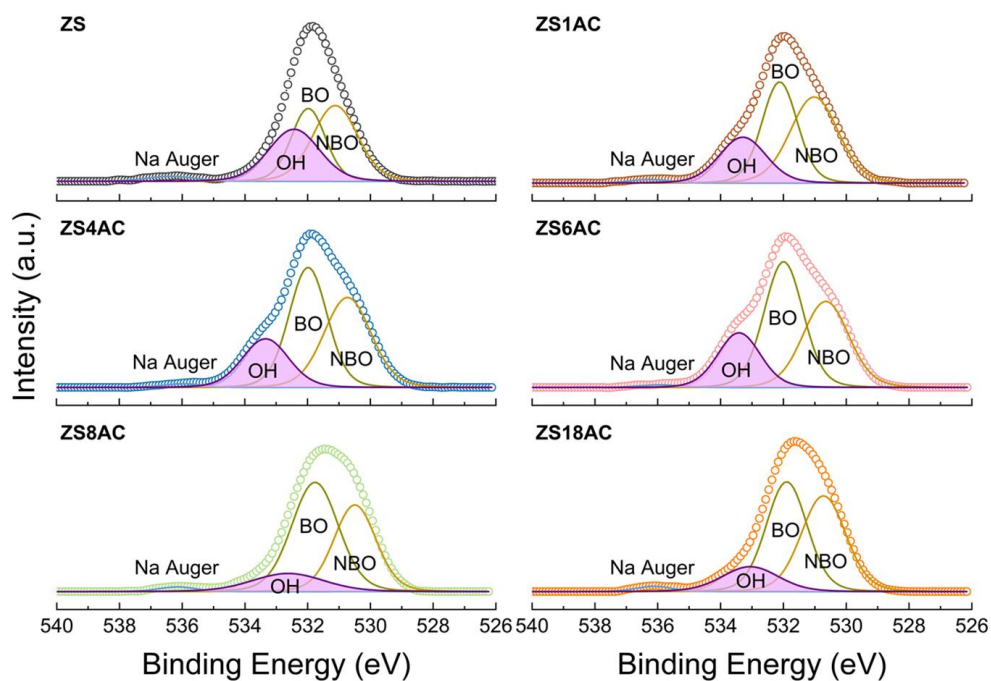


Figure 3.7. Deconvoluted O1s XPS spectra of ZS and ZS#ACs.

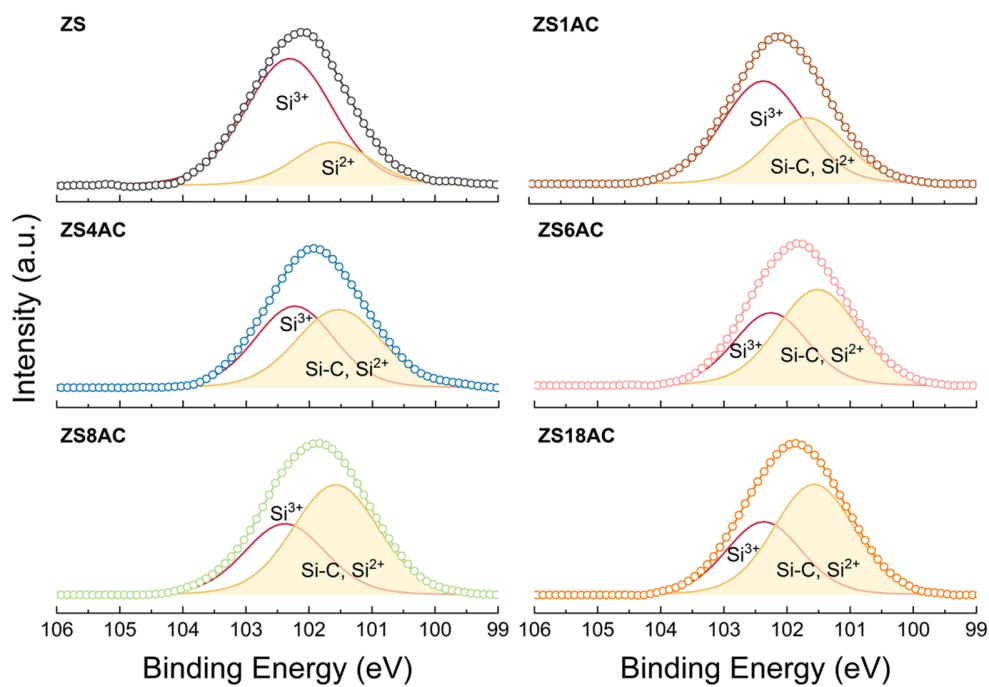


Figure 3.8. Deconvoluted Si 2p XPS spectra of ZS and ZS#ACs.

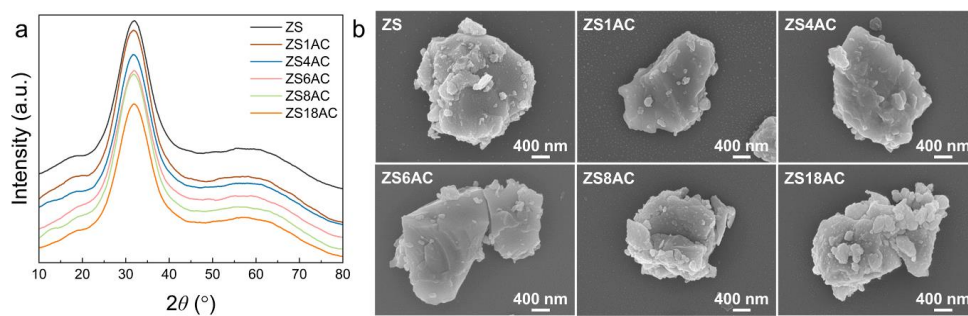


Figure 3.9. (a) XRD patterns and (b) SEM images of ZS and ZS#ACs.

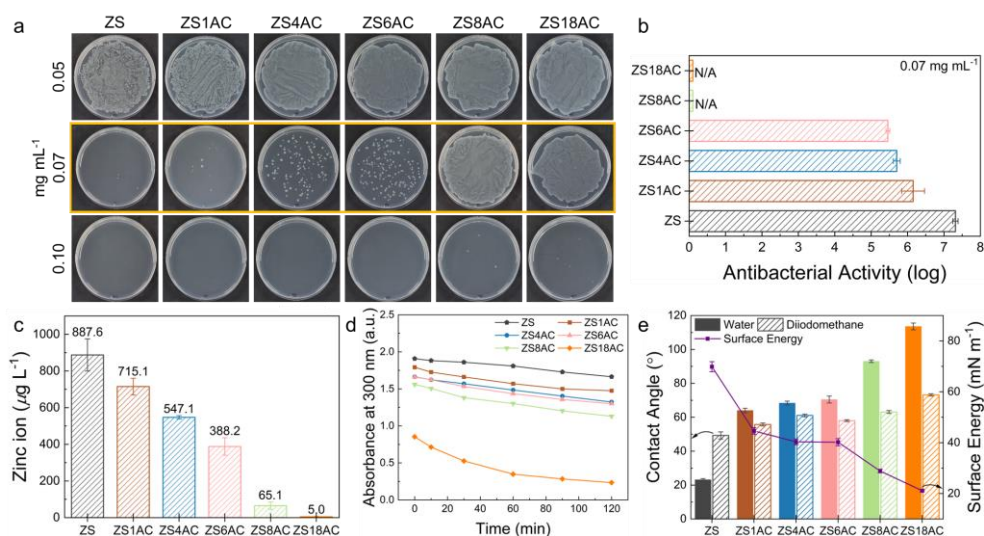


Figure 3.10. (a) Photographic results demonstrating antibacterial activity of ZS and ZS#ACs. (b) The calculated antibacterial activity of ZS and ZS#ACs against *E. coli* at a suspension concentration of 0.07 mg mL⁻¹. (c) The amount of released zinc ions from ZS and ZS#ACs measured by ICP-MS. (d) Time-dependent absorbance of ZS and ZS#ACs suspensions at 300 nm. (e) Contact angle values of water and diiodomethane for ZS and ZS#ACs and their surface energy calculated using the Owens–Wendt–Rabel–Kaelble (OWRK) method.

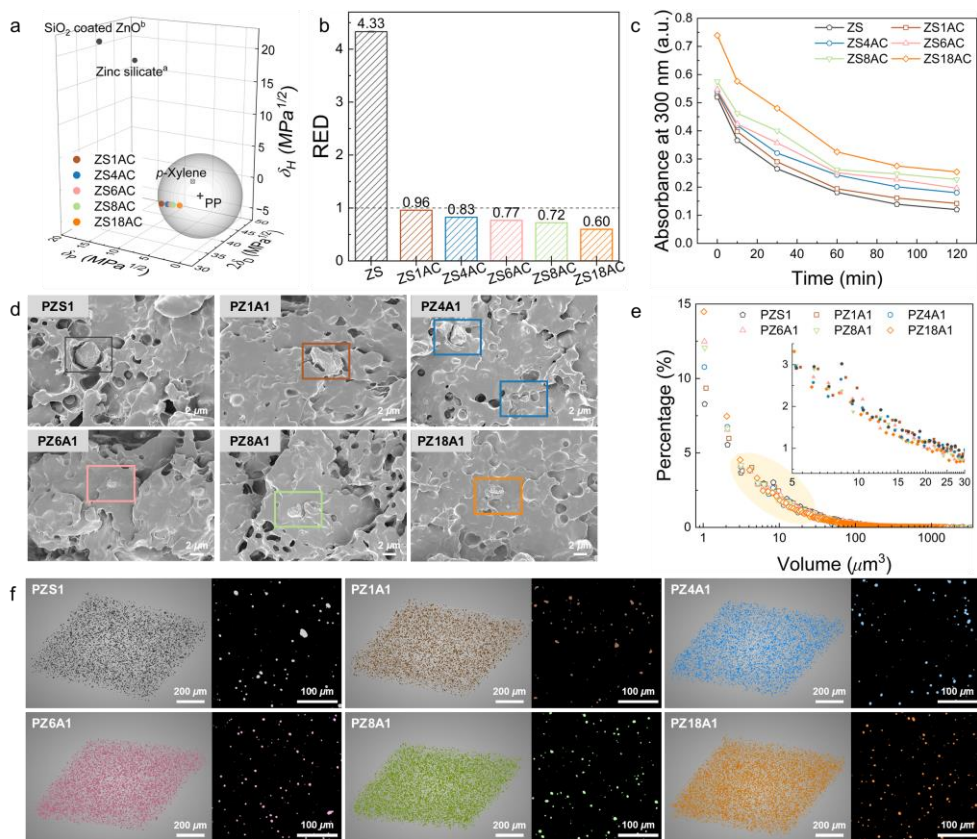


Figure 3.11. (a) Hansen solubility parameters (HSP) of PP, *p*-xylene, ZS#ACs and ZS analogues. (b) Calculated relative energy difference (RED) values for ZS and ZS#ACs. The RED value of ZS was calculated using HSP values of its analogues. (c) Time-dependent absorbance value at 300 nm of ZS and ZS#ACs suspensions in *p*-xylene. (d) Cross-sectional SEM images of fractured PZS1 and PZ#A1s. (e) Particle volume distributions in PZS1 and PZ#A1s determined by XRM. (f) 3D (left) and 2D (right) XRM images of PZS1 and PZ#A1s.

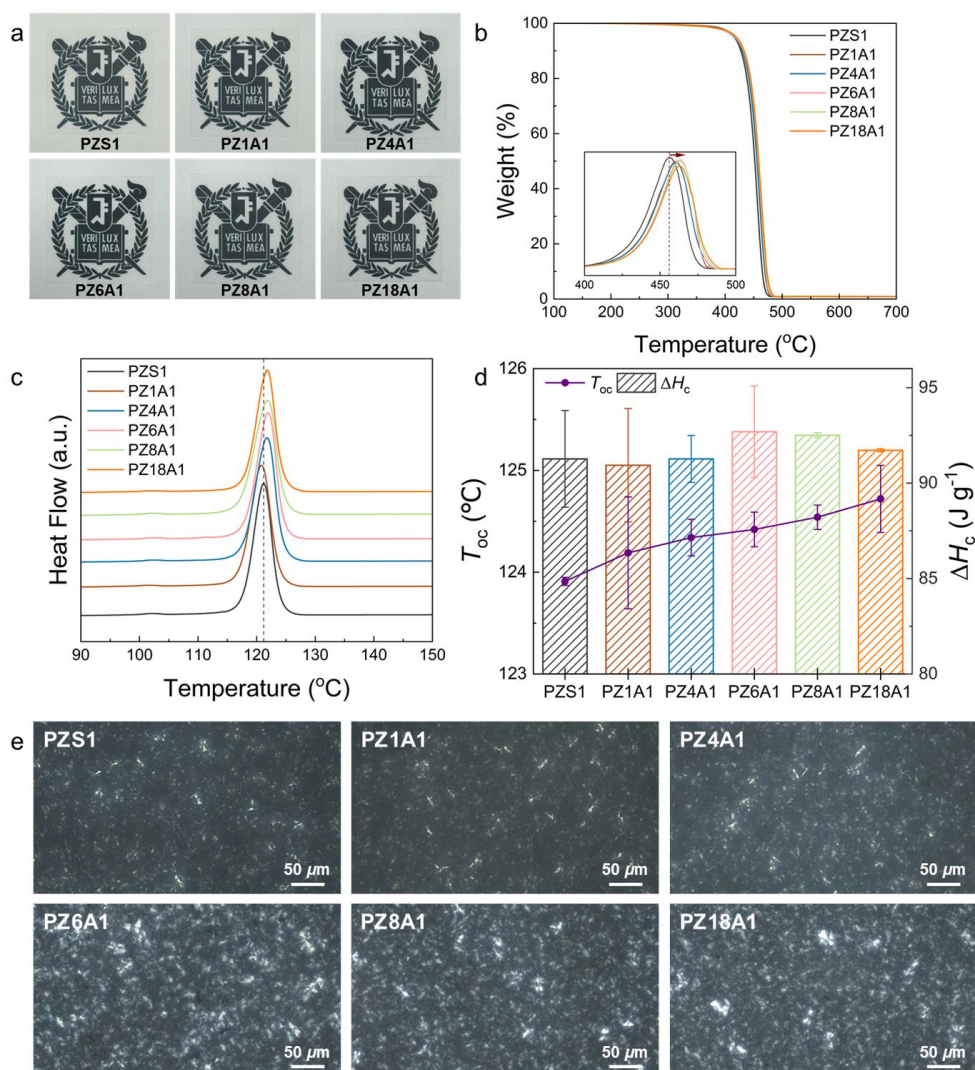


Figure 3.12. (a) Photographs of PZS1 and PZ#A1s. (b) TGA curves of PZS1 and PZ#A1s. (inset: 1st derivatives of TGA curves) (c) DSC cooling curves of PZS1 and PZ#A1s. (d) T_{oc} and ΔH_c of PZS1 and PZ#A1s. (e) POM images of PZS1 and PZ#A1s.

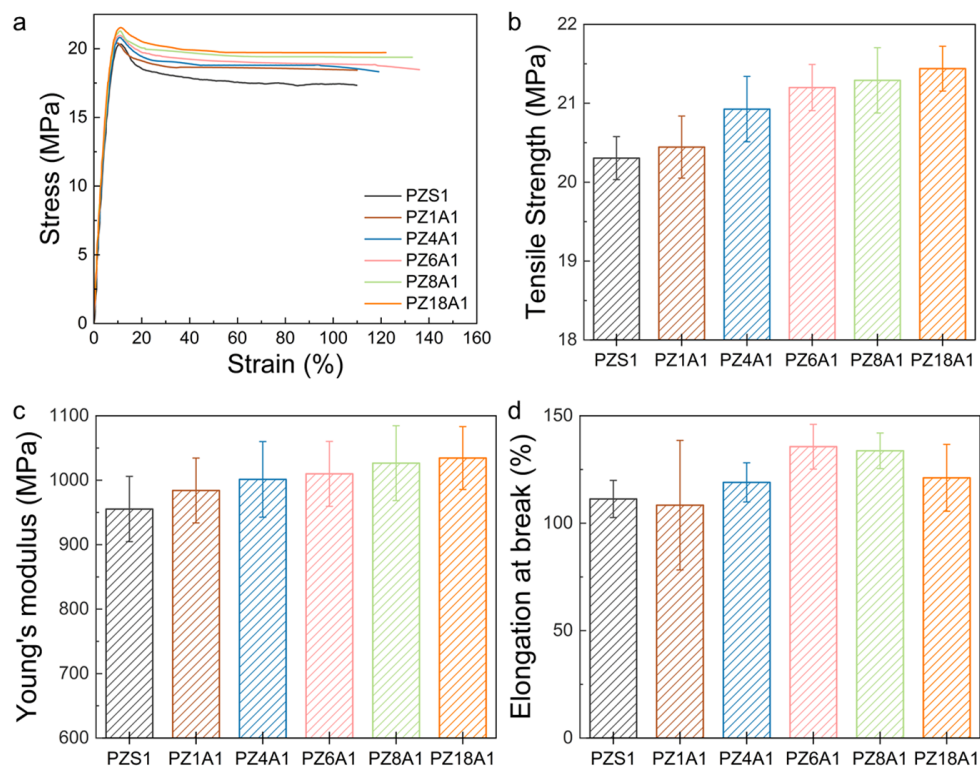


Figure 3.13. (a) Stress-strain curves of PZS1 and PZ#A1s. (b) Tensile strength, (d) Young's modulus, and (d) elongation at break of PZS1, and PZ#A1s.

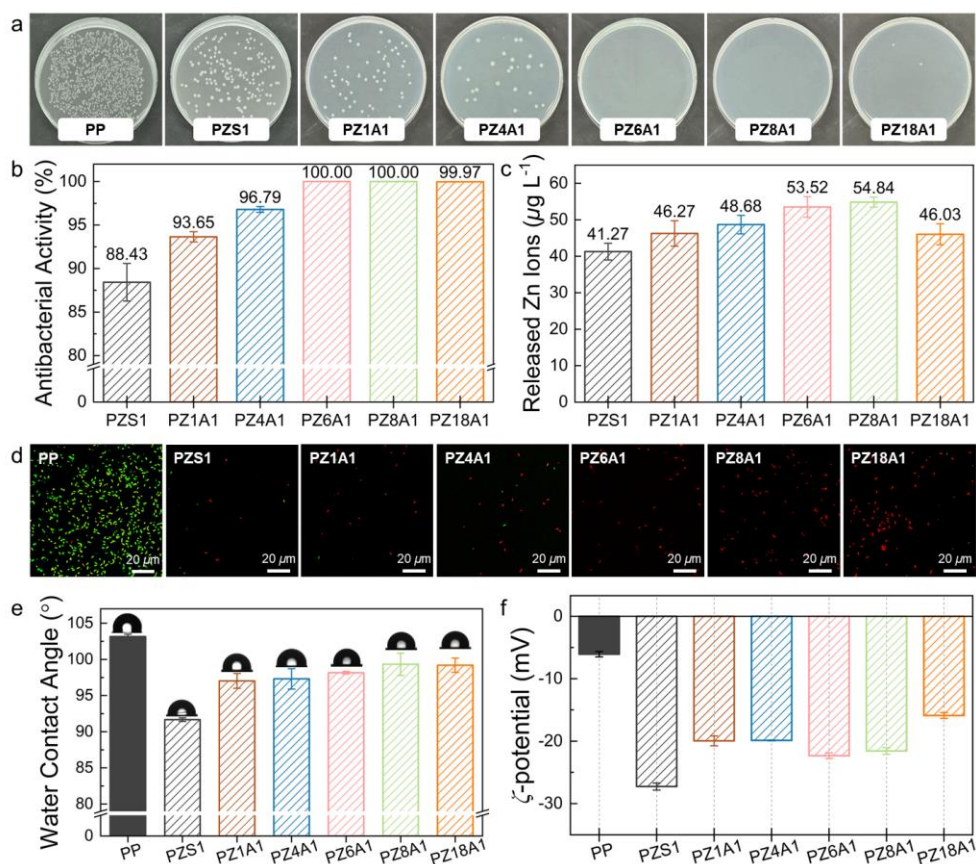


Figure 3.14. (a) Photographic results demonstrating the antibacterial activity of PZS1 and PZ#A1s against *E. coli*. (b) Calculated antibacterial activity of PZS1 and PZ#A1s. (c) The amount of released zinc ions from PZS1 and PZ#A1s measured by ICP-MS. (d) CLSM images, (e) Water contact angle values, and (f) surface ζ -potential of PP, PZS1, and PZ#A1s.

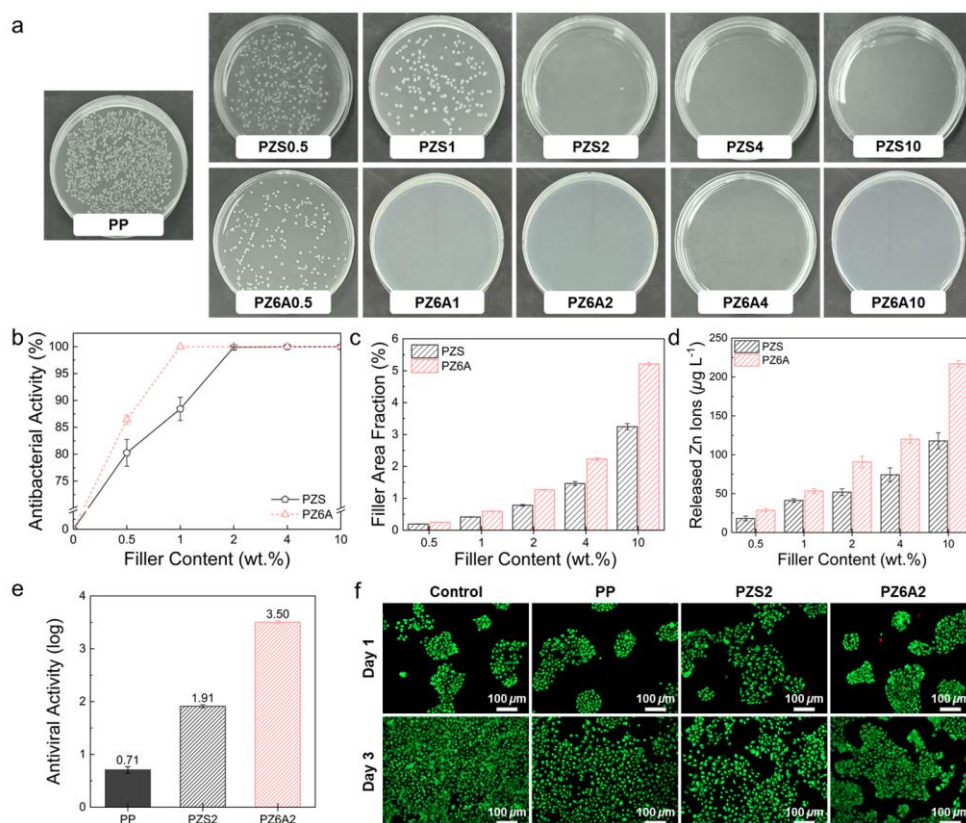


Figure 3.15. (a) Photographic results demonstrating the antibacterial activity of PZSx and PZ6Ax series against *E. coli*. (b) Calculated antibacterial activity of PZSxs and PZ6Axs. (c) Filler area fraction on the surface of PZSxs and PZ6Axs determined by BSE mode of FE-SEM. (d) The amount of released zinc ions from PZSxs and PZ6Axs measured by ICP-MS. (e) Antiviral activity against MS2 coliphage of PP, PZSxs, and PZ6Axs. (f) CLSM images of HaCaT cells cultured with TCP (Tissue culture grade polystyrene, control), PP, PZSxs, and PZ6Axs for 1 day and 3 days.

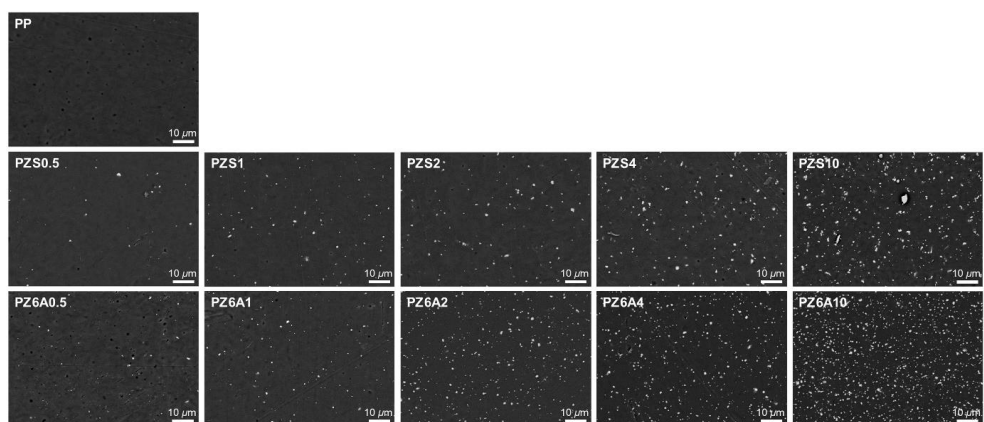


Figure 3.16. SEM-BSE images of PP, PZSxs, and PZ6Axs.

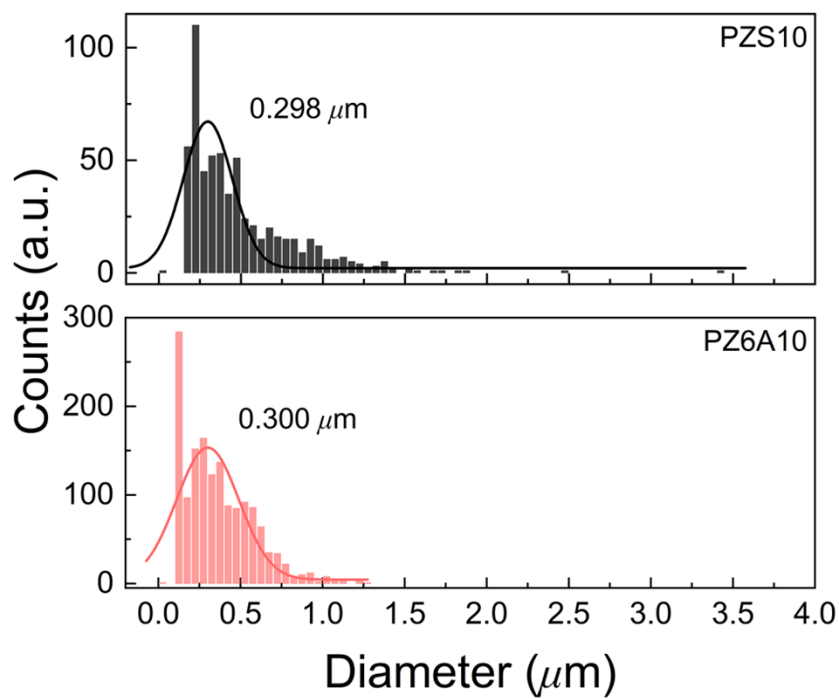


Figure 3.17. Size distribution of fillers on the surface of PZS10 and PZ6A10, determined by BSE images.

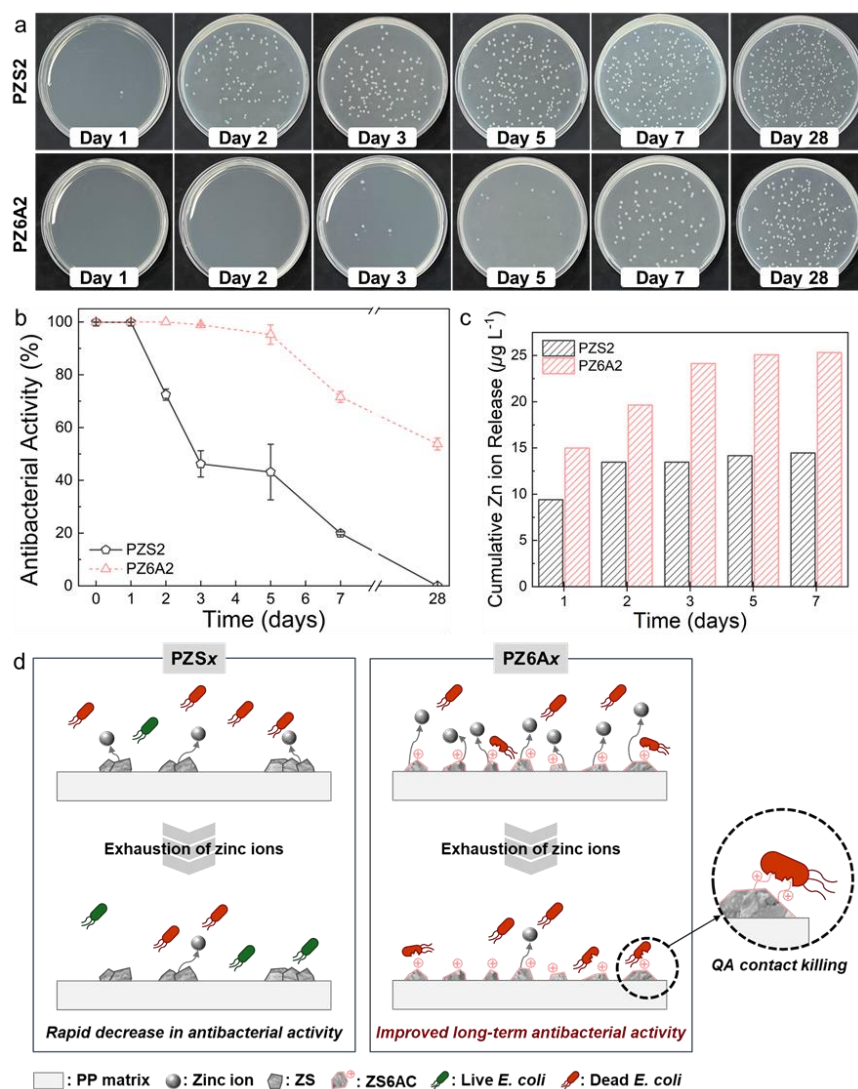


Figure 3.18. (a) Photographic results demonstrating antibacterial activity of PZS2 and PZ6A2 after immersion in distilled water for various durations. (b) Calculated antibacterial activity of PZS2 and PZ6A2. (c) Time-dependent cumulative release of zinc ions from PZS2 and PZ6A2 measured by ICP-MS. (d) Proposed dual antibacterial mechanisms of PZ6A2 enhancing long-term antibacterial activity.

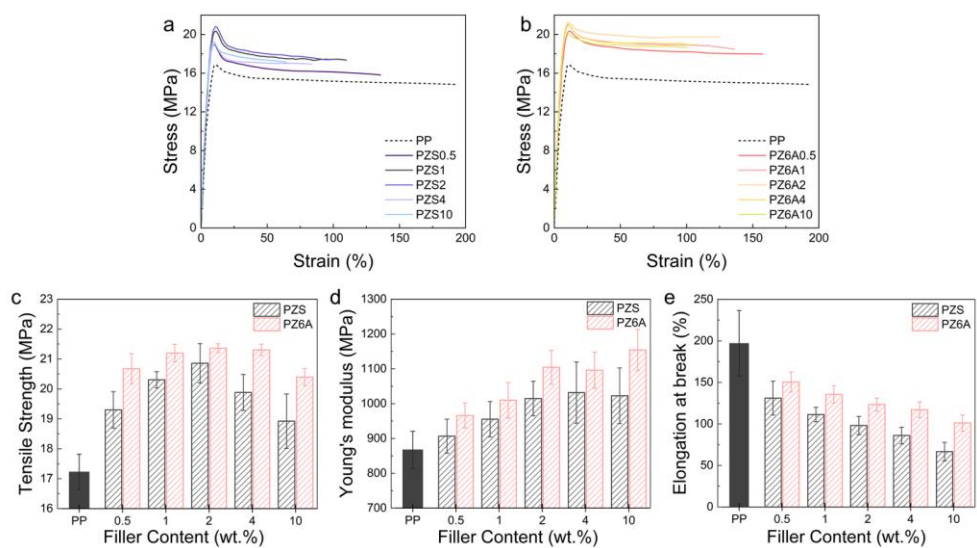


Figure 3.19. Stress-strain curves of (a) PZSxs and (b) PZ6Axs. (c) Tensile strength, (d) Young's modulus, and (e) elongation at break values of PP, PZSxs, and PZ6Axs.

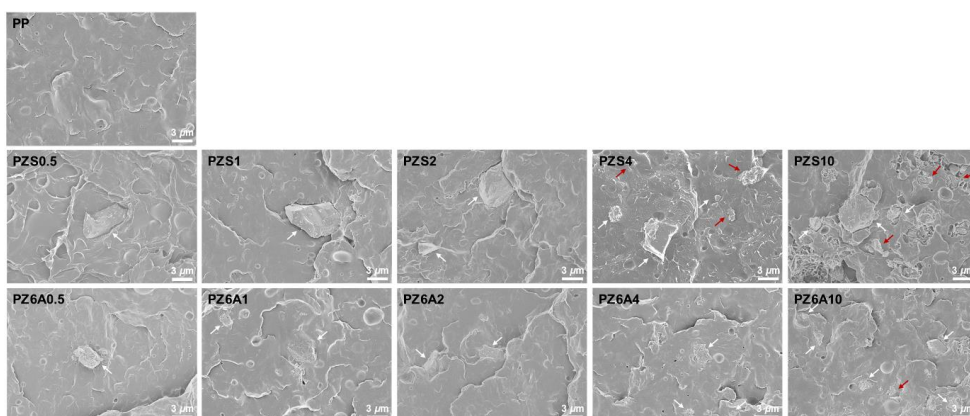


Figure 3.20. Cross-sectional SEM images of PP, PZSxs, and PZ6Axs. The white arrow indicates the filler, and the red arrow indicates the aggregate.

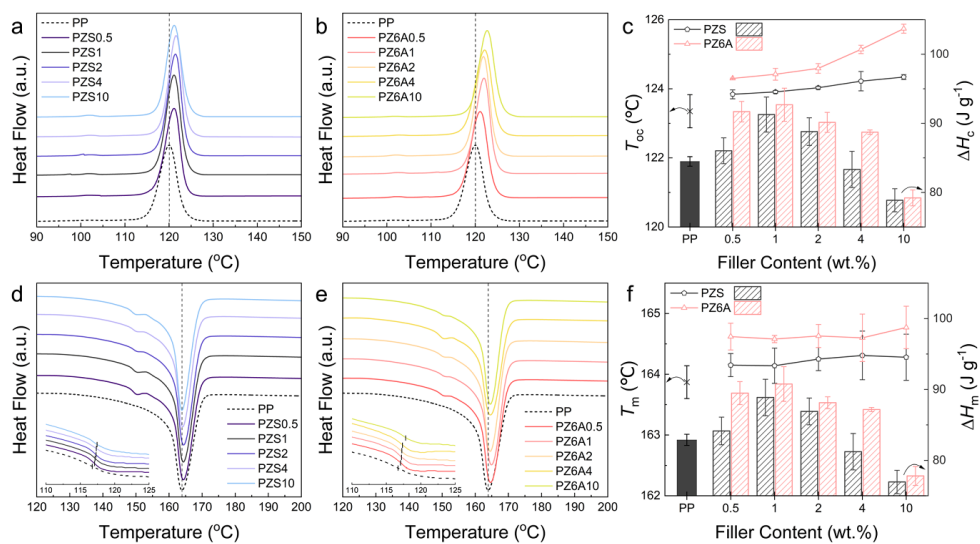


Figure 3.21. DSC cooling curves of (a) PZSxs and (b) PZ6Axs. (c) T_{oc} and ΔH_c values for PP, PZSxs, and PZ6Axs. 2nd heating curves of (d) PZSxs and (e) PZ6Axs. (f) T_m and ΔH_m values for PP, PZSxs, and PZ6Axs.

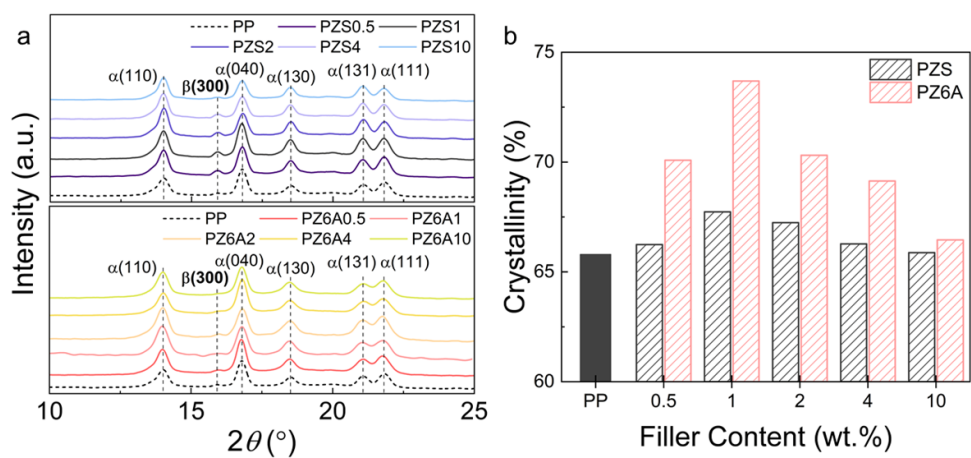


Figure 3.22. (a) XRD patterns of PP, PZSxs, and PZ6Axs. (b) Crystallinity of PP, PZSxs, and PZ6Axs determined by XRD data.

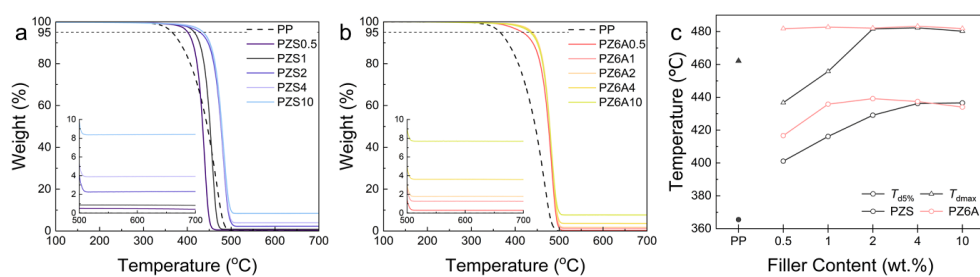


Figure 3.23. TGA curves of (a) PZSxs and (b) PZ6Axs. (c) $T_{d5\%}$ and T_{dmax} of PP, PZSxs, and PZ6Axs.

초 록

본 논문은 고분자 매트릭스와 높은 상용성을 갖는 동시에 강력한 항균성을 가지는 필러를 개발하고, 이를 폴리아마이드 박막 나노복합막과 폴리프로필렌 복합재료에 도입함으로써 각 고분자 복합재료의 증가된 항균성 및 다양한 특성 변화에 관해 기술하였다. 첫째, 셀룰로오스 나노결정을 카다놀 유래 물질인 3-펜타데실페놀 (3-PDP)을 이용해 개질하여 양친매성 기능화 셀룰로오스 나노결정(f-CNC)를 합성하고, 이를 폴리아마이드막을 형성하는 계면중합 과정에서 유기용매에 나노필러로 분산시켜 폴리아마이드-f-CNC 복합재료막을 제작하였다. 도입된 f-CNC는 계면중합 과정에서 단 분자의 확산 과정을 조절하고 물리적으로 막의 과도한 성장이 일어나지 않도록 억제함으로써, 계면 사이에서 형성되는 폴리아마이드의 두께와 투과 저항을 크게 낮추어 막의 수투과도를 향상시켰다. f-CNC 도입 이후에도 폴리아마이드 막은 염제거율의 저하를 보이지 않았고, 9일간의 장기 구동에서도 수투과도와 염제거율이 유지되었다. 이를 통해, 응집된 필러에 의한 분리 성능 저하 없이 f-CNC가 폴리아마이드 매트릭스 내에 고르게 분포되어 있음을 확인할 수 있었다. 또한, 날카롭고 단단한 구조를 이용하여 세포벽을

허무는 효과를 갖는 셀룰로오스 나노결정과 긴 탄소사슬을 특징으로 하여 세균에 대해 접촉-살균 효과를 가지는 3-PDP가 동시에 도입됨에 따라 폴리아마이드-f-CNC 복합재료막의 표면은 강력한 항균성을 갖게 되었다. 또한, 원료 용액에 세균이나 유기 오염물질을 도입하고 막 표면에서 일어나는 오염 정도에 대해서도 평가를 진행하였는데 폴리아마이드-f-CNC 복합재료막은 기존 폴리아마이드막에 비해서 오염 과정에서 수투과도의 감소가 더 적게 일어날 뿐만 아니라, 세척 과정을 통한 수투과도의 회복률 역시 더 높은 것이 확인되었다. 종합하자면, f-CNC의 도입은 폴리아마이드 복합재료막의 투과도, 항균성, 방오성 등을 동시에 향상시켰다.

둘째, 아연 이온의 용출을 통해 항균 활성을 갖는 아연 규산염(ZS)을 탄소 개수가 #인 탄소 사슬과 사차암모늄 작용기를 가지는 실레인 커플링 에이전트를 이용해 개질함으로써 ZS#AC를 합성하고, 이를 폴리프로필렌에 도입함으로써 PZ#A_x (x는 폴리프로필렌 대비 도입된 필러의 질량 비율을 나타냄) 복합재료를 제작하였다. ZS#AC의 탄소 사슬은 폴리프로필렌 매트릭스 내에서의 필러의 분산성을 향상시킴으로써, ZS의 아연 이온 용출이 폴리프로필렌 복합재료에서도 원활히 일어날 수 있게 하여 항균성을 증가시켰다. 도입된 사차암모늄 작용기는 항균 활성을 더 증가시키

는 동시에, 아연 이온의 용출 이후에도 접촉-살균 효과에 의한 항균 활성이 지속될 수 있도록 하였다. 또한, 단단한 특성과 폴리프로필렌 매트릭스에 대한 높은 분산성을 동시에 갖는 ZS#AC의 도입은 폴리프로필렌의 기계적 특성 및 열적 특성 향상에도 도움을 주었다. 종합하자면, ZS#AC의 도입은 폴리프로필렌 복합재료의 항균성, 물성, 열적 안정성 등을 동시에 향상시켰다.

주요어: 고분자 복합재료, 항균성 필러, 폴리아마이드 박막 나노복합막, 폴리프로필렌 복합재료, 셀룰로오스 나노결정, 아연 규산염, 필러 개질, 필러 분산성

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